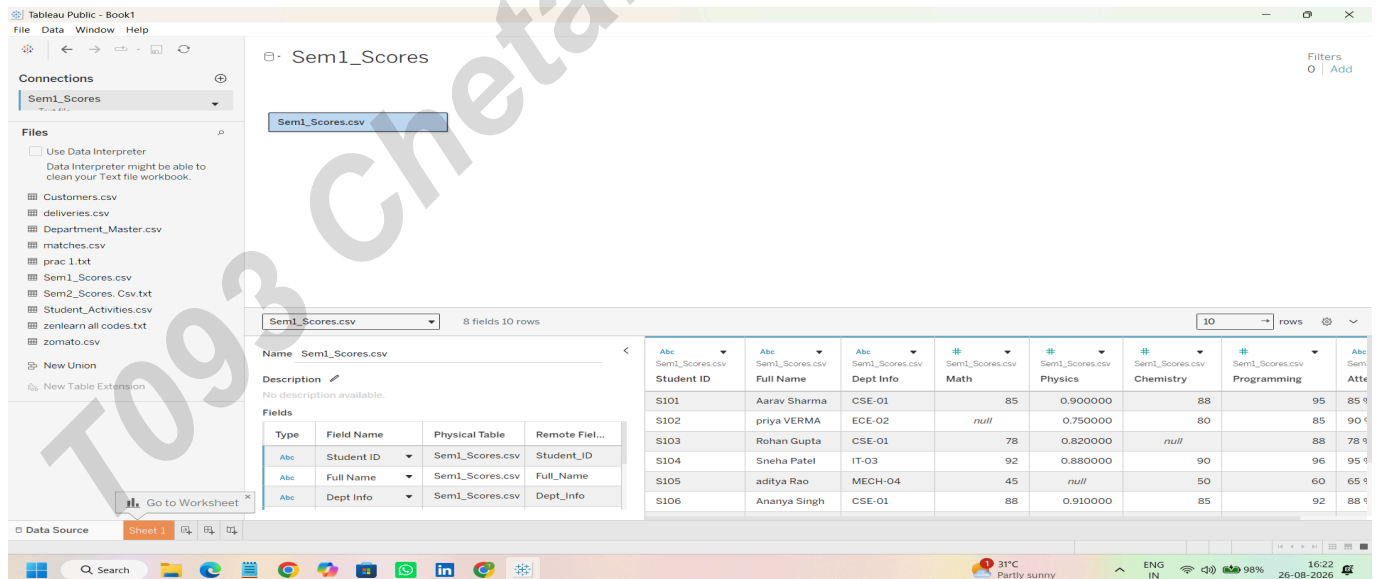
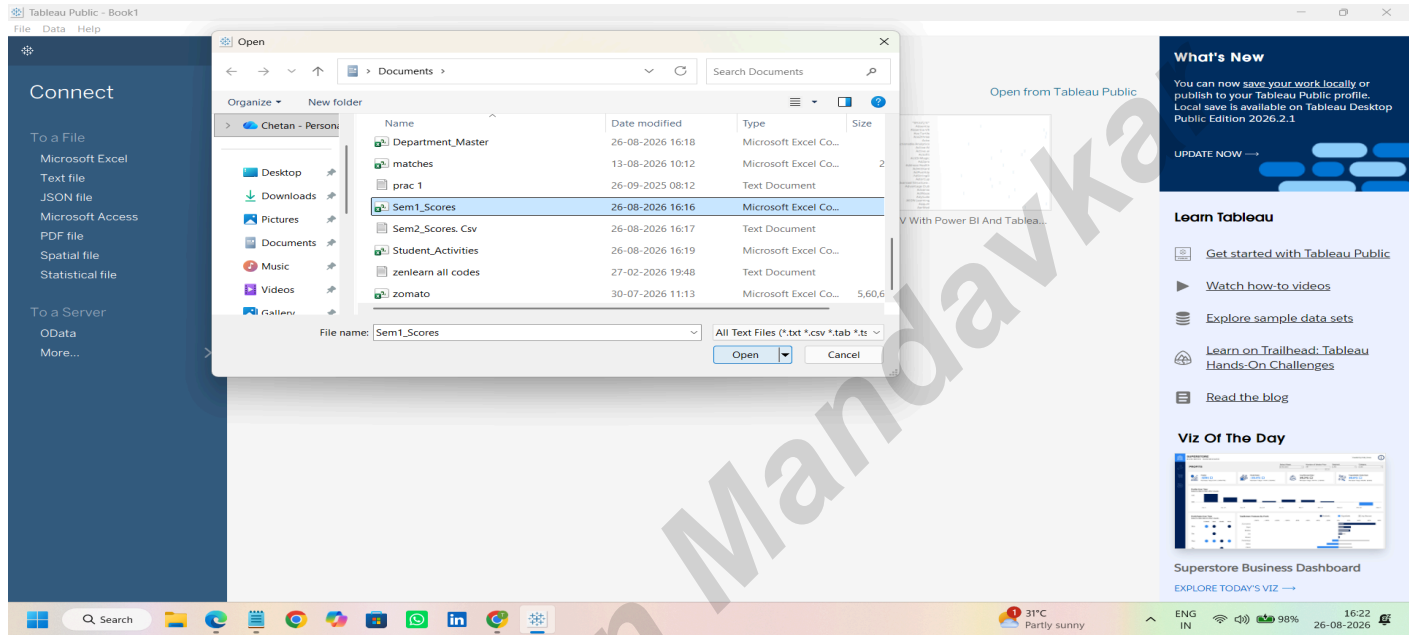


Sheth L.U.J. And Sir M.V. College

Aim:- Cleaning and Structuring Data for Analysis

A: Identify and Handle Missing or Inconsistent Data

Step 1:- Open Tableau And Select The CSV File And Select OK



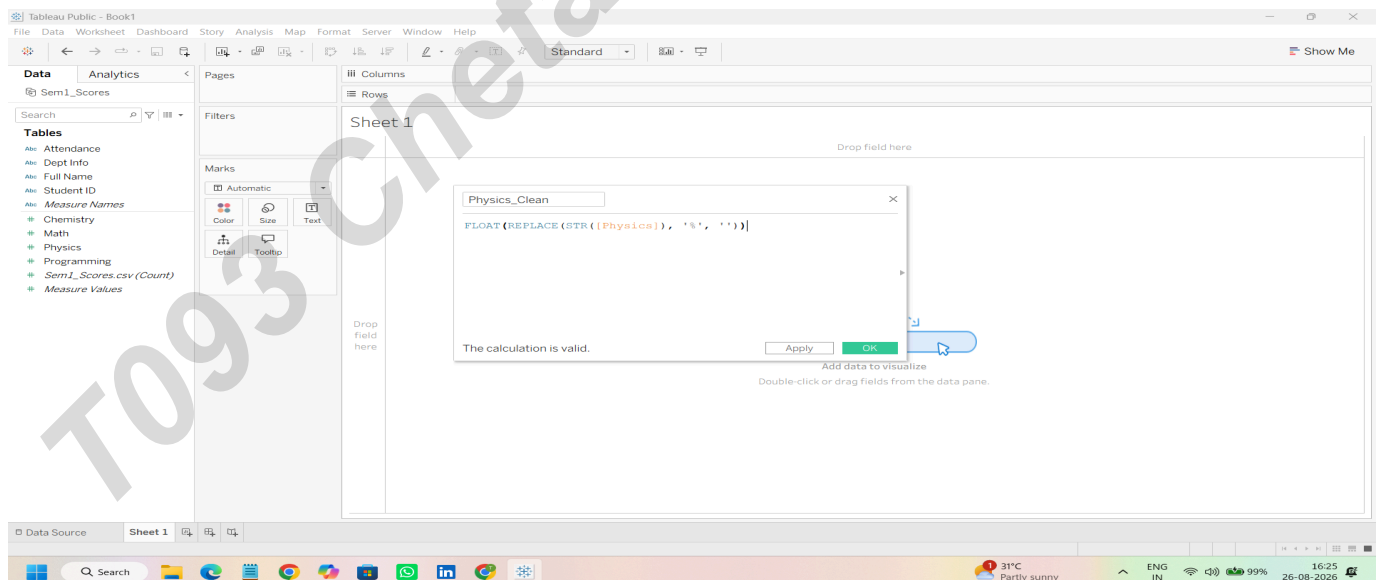
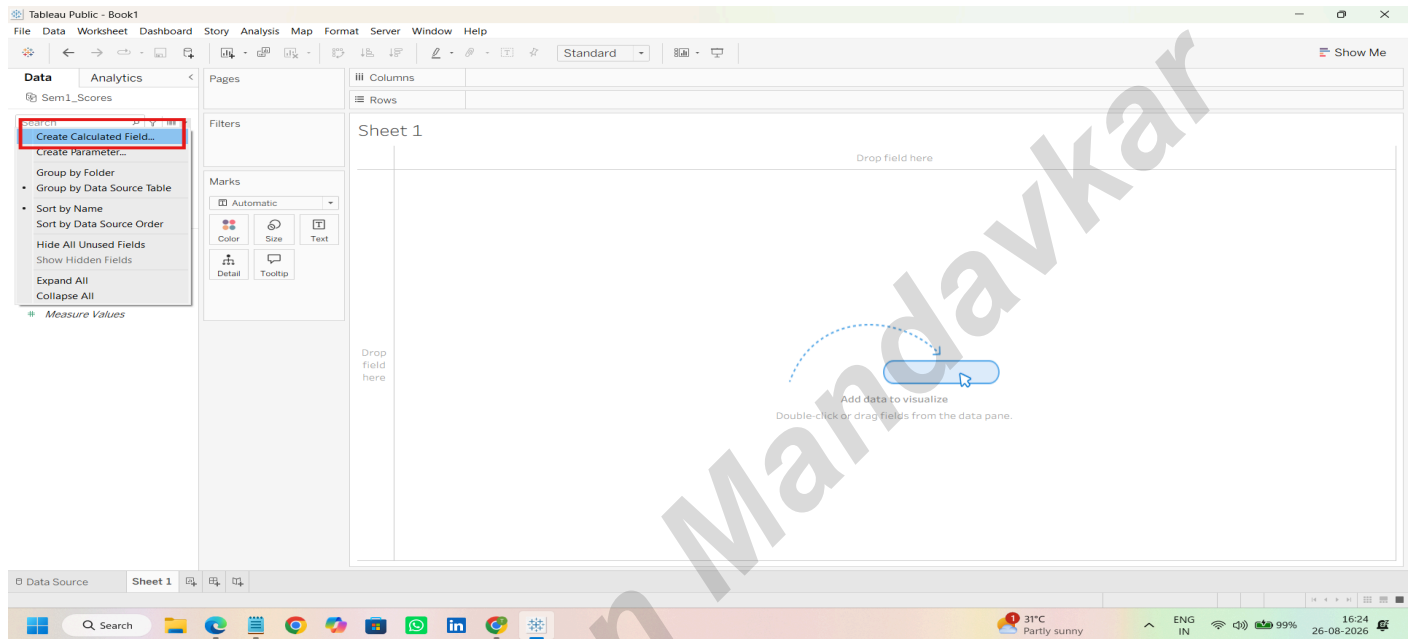
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

Sheth L.U.J. And Sir M.V. College

Step 2:- Navigate to Sheet 1 (bottom-left tab). Right-click an empty space in the Data Pane (left sidebar) --> select Create Calculated Field.... Create the cleaned measure for Physics: o Field Name: Physics_Clean o Tableau Formula: Code snippet `FLOAT(REPLACE(STR([Physics])), '%', ''))` o Click OK.



Name:- Chetan Mandavkar

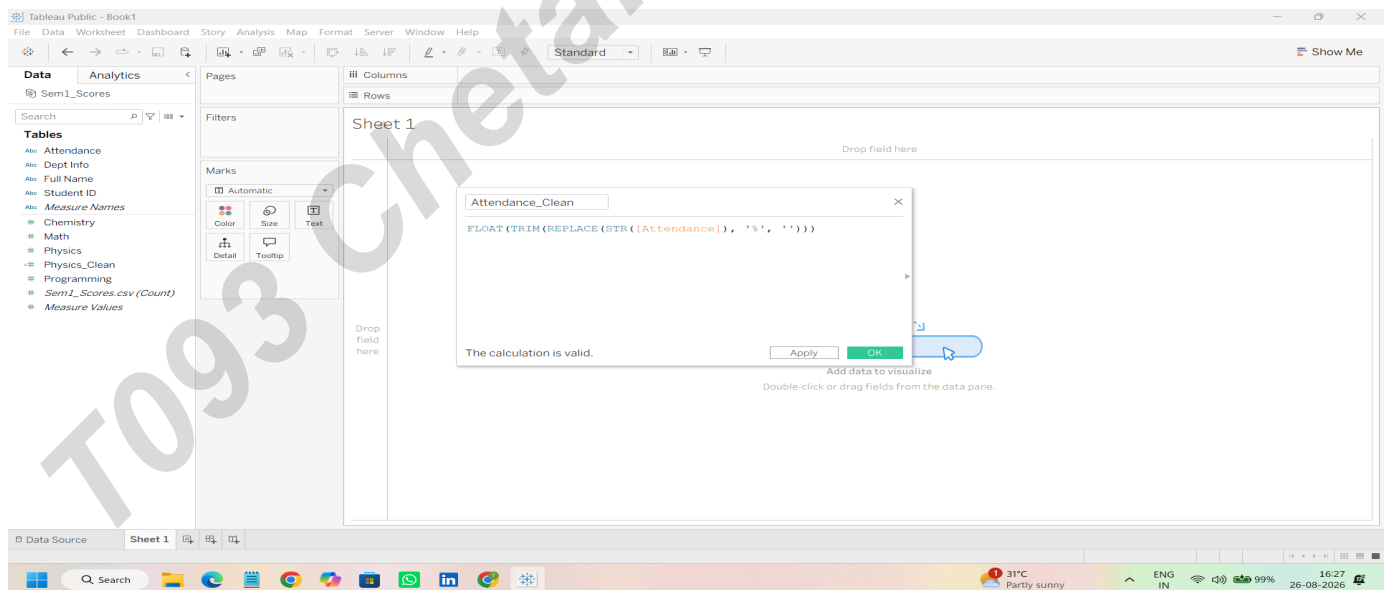
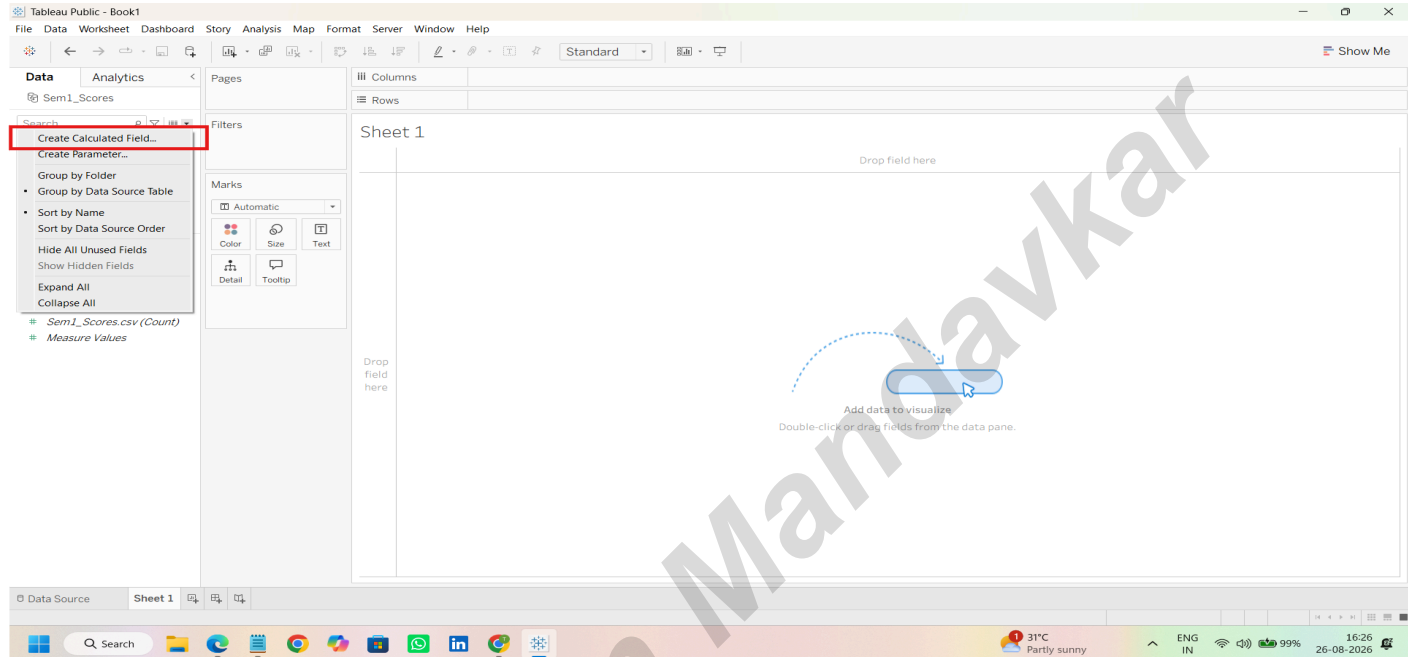
Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

Sheth L.U.J. And Sir M.V. College

Step 3:- Create the cleaned measure for Attendance:

- o Right-click the Data Pane --> Create Calculated Field....
- o Field Name: Attendance_Clean
- o Tableau Formula: Code snippet `FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`
- o Click OK.



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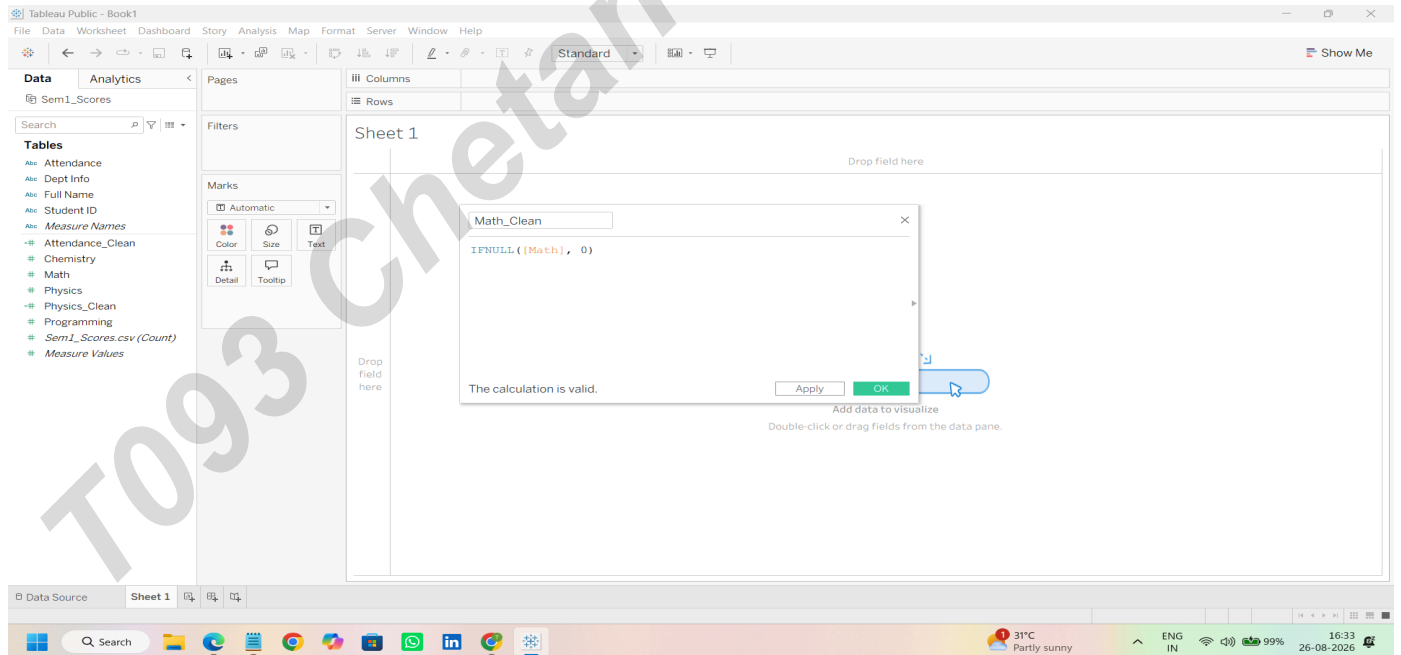
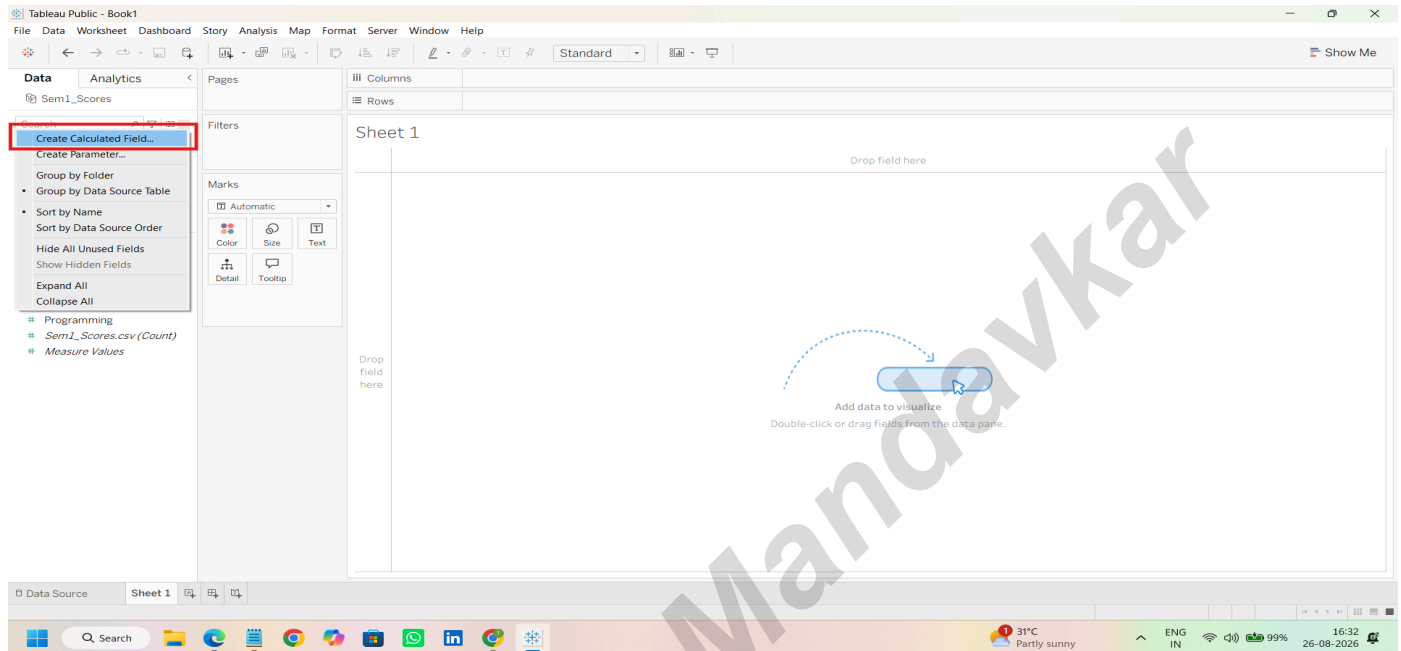
Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

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Step 4:- Handle missing entries in Math:

- o Right-click the Data Pane
- > **Create Calculated Field....**
- o Field Name: **Math_Clean**
- o Tableau Formula: **Code snippet IFNULL([Math], 0)**
- o Click OK.



Name:- Chetan Mandavkar

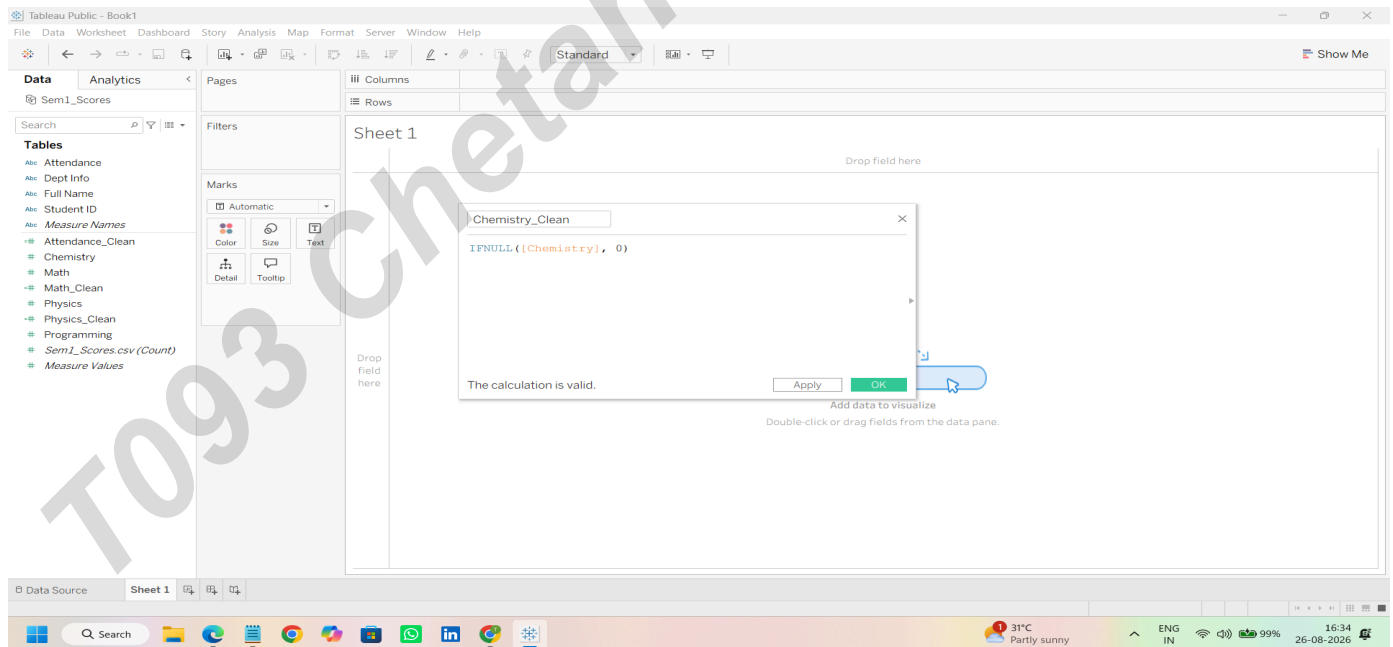
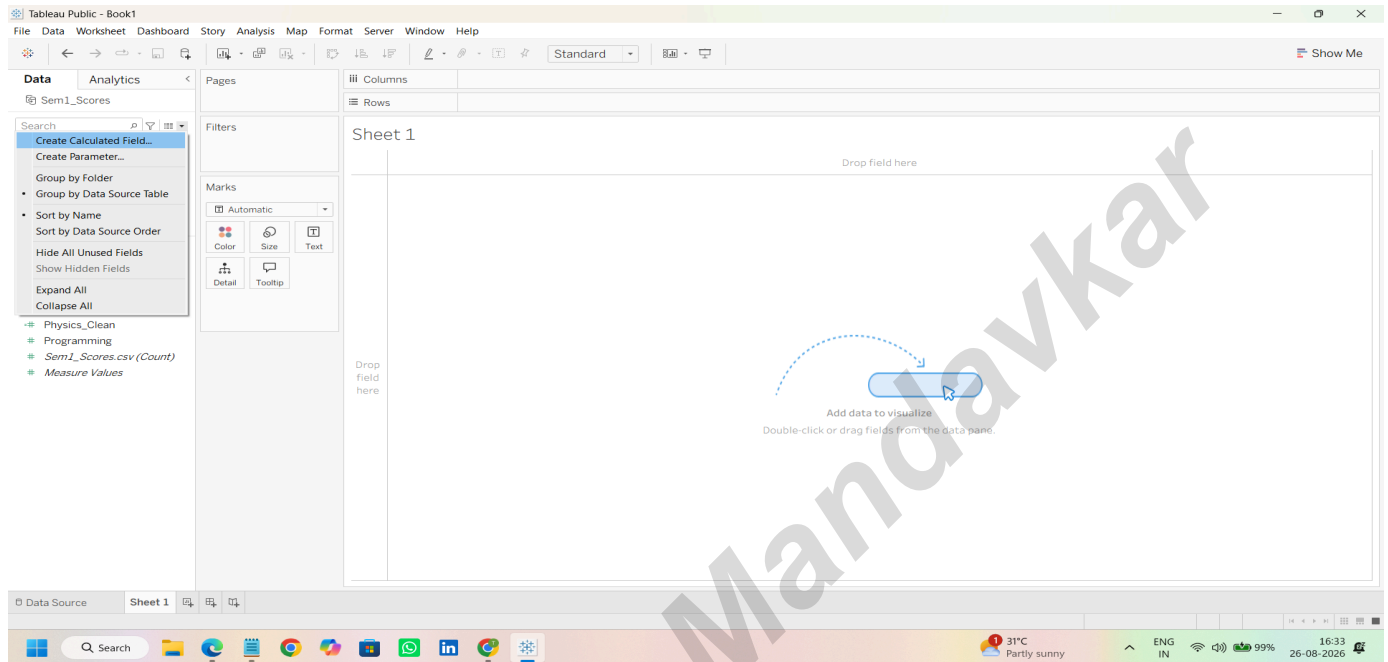
Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

Sheth L.U.J. And Sir M.V. College

Step 5:- Handle missing entries in Chemistry:

- o Right-click the Data Pane --> Create Calculated Field....
- o Field Name: Chemistry_Clean
- o Tableau Formula: Code snippet IFNULL([Chemistry], 0)
- o Click OK.



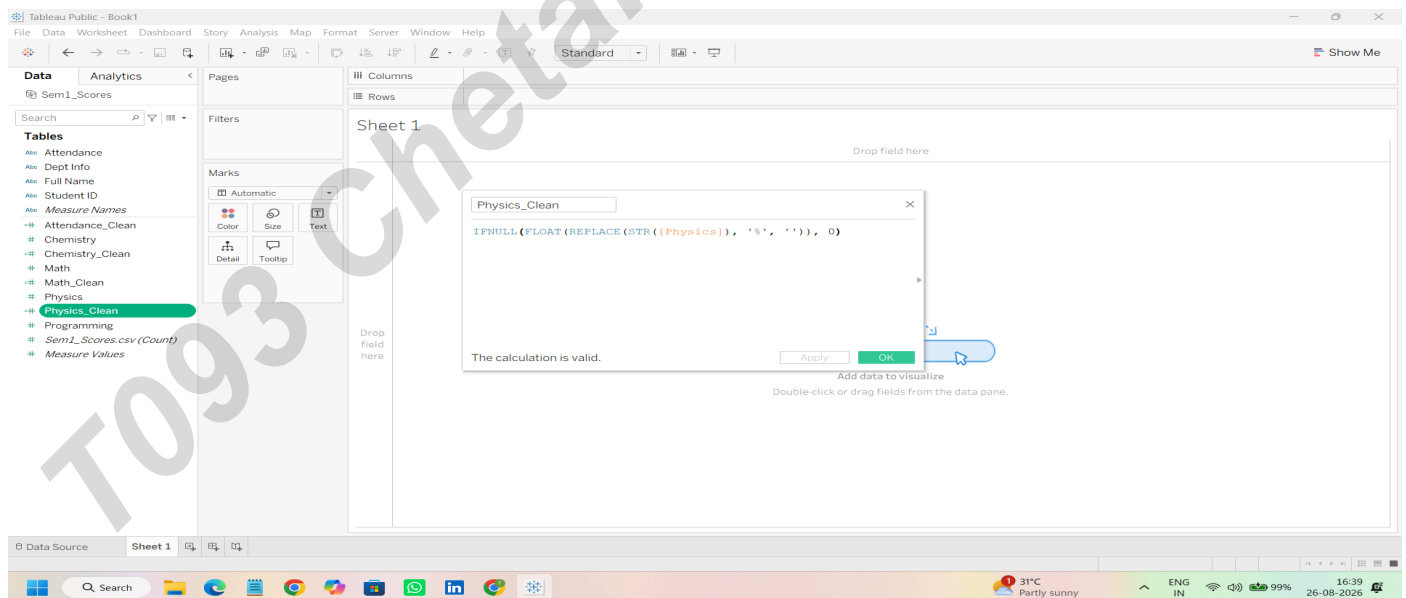
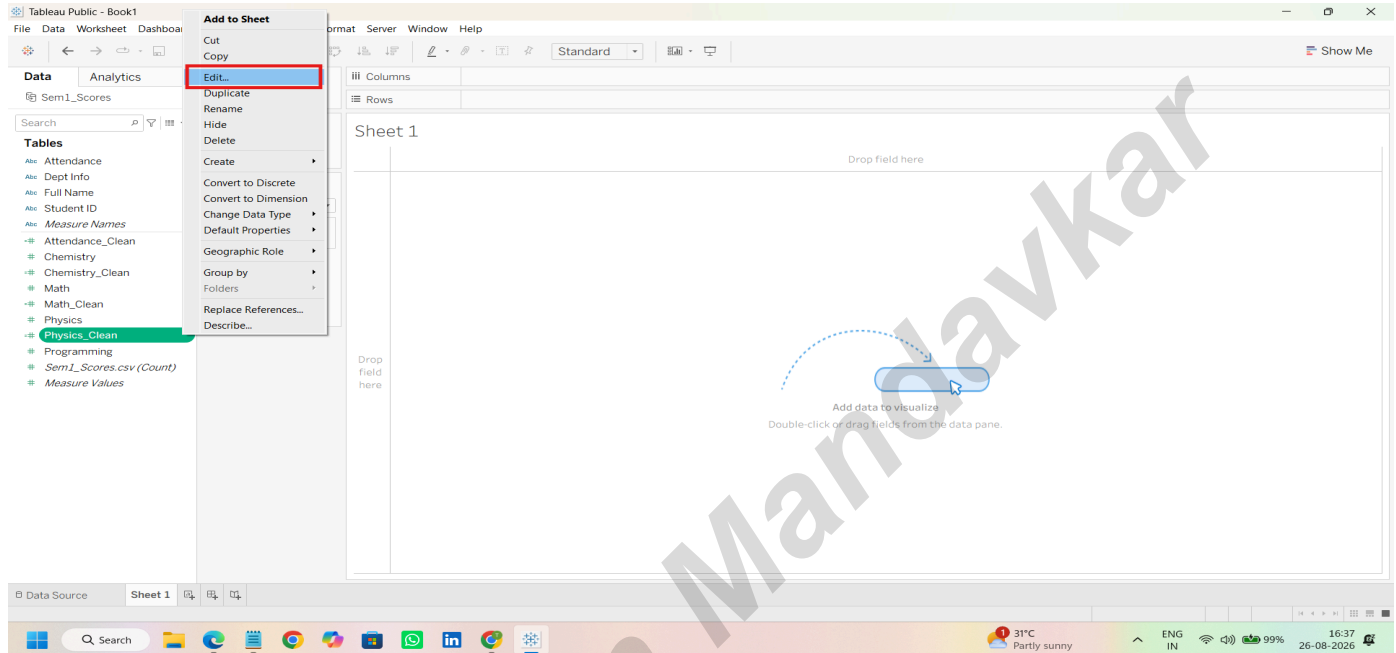
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

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Step 6:- Student S105 contains a Null value in Physics. Combine string stripping with null handling: o Edit Physics_Clean (right-click Physics_Clean --> Edit...): o Updated Tableau Formula: Code snippet IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0) o Click OK.



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Subject:- Data Visualization With Power BI And Tableau

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B: Convert Wide Data into Tall Format

Step 1:- In Tableau, navigate to the Data Source tab at the bottom-left corner. 2. Locate the four horizontal subject score columns: o Math o Physics o Chemistry o Programming

The screenshot shows the Tableau Public interface with the 'Sem1_Scores' data source loaded. The 'Fields' pane on the left is highlighted with a red box, showing a list of fields: Math, Physics, Chemistry, Programming, and Attendance. The main view displays a table with columns for Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, and Attendance. The Math, Physics, Chemistry, and Programming columns are highlighted in blue.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85
S102	priya VERMA	ECE-02	null	0.750000	80	85	90
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78
S104	Sneha Patel	IT-03	92	0.880000	90	96	95
S105	aditya Rao	MECH-04	45	null	50	60	65
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98

Step 2:- Click the header of the Math column. Hold down Ctrl (or Cmd on Mac) and click Physics, Chemistry, and Programming so all four columns are highlighted in blue. Right-click on any of the highlighted column headers. Select Pivot.

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Subject:- Data Visualization With Power BI And Tableau

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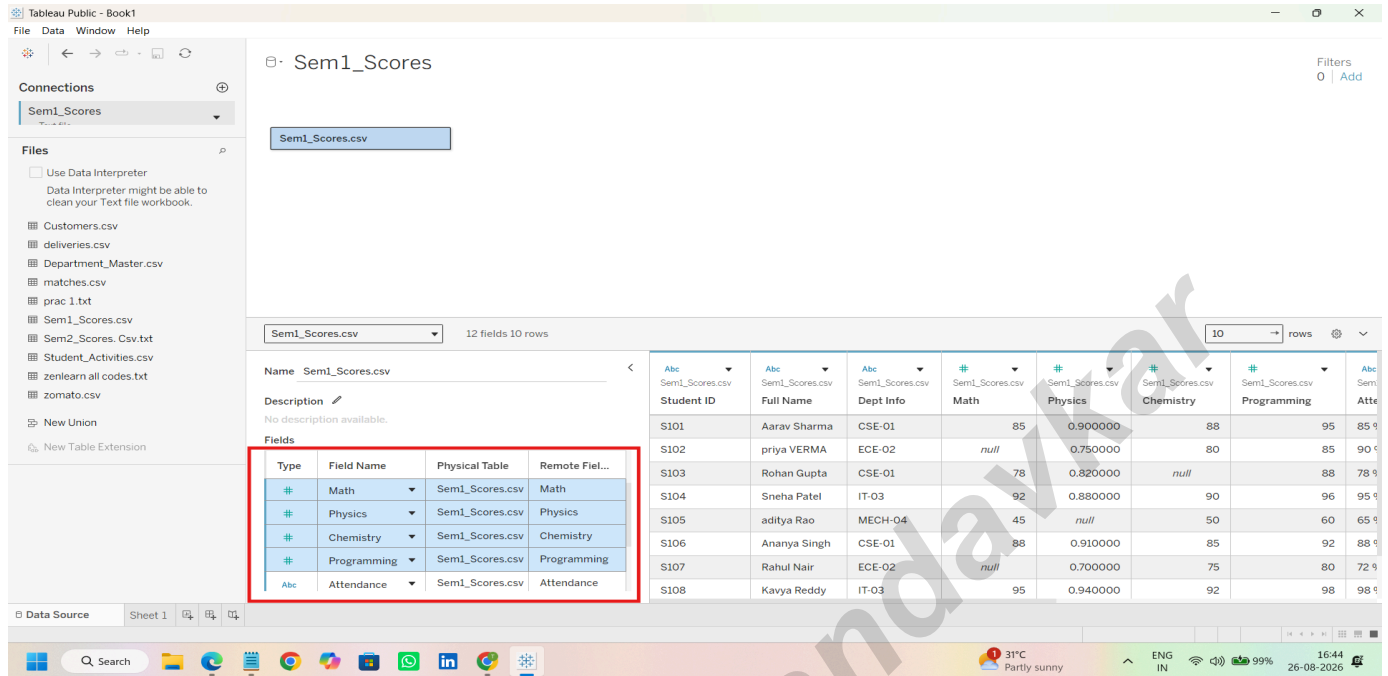


Tableau Public - Book1

File Data Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Customers.csv
deliveries.csv
Department_Master.csv
matches.csv
prac 1.txt
Sem1_Scores.csv
Sem2_Scores.Csv.txt
Student_Activities.csv
zenlearn all codes.txt
zomato.csv

New Union
New Table Extension

Sem1_Scores

Sem1_Scores.csv

12 fields 10 rows

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
#	Math	Sem1_Scores.csv	Math
#	Physics	Sem1_Scores.csv	Physics
#	Chemistry	Sem1_Scores.csv	Chemistry
#	Programming	Sem1_Scores.csv	Programming
Abc	Attendance	Sem1_Scores.csv	Attendance

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85%
S102	priya VERMA	ECE-02	null	0.750000	80	85	90%
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78%
S104	Sneha Patel	IT-03	92	0.880000	90	96	95%
S105	aditya Rao	MECH-04	45	null	50	60	65%
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88%
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72%
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98%

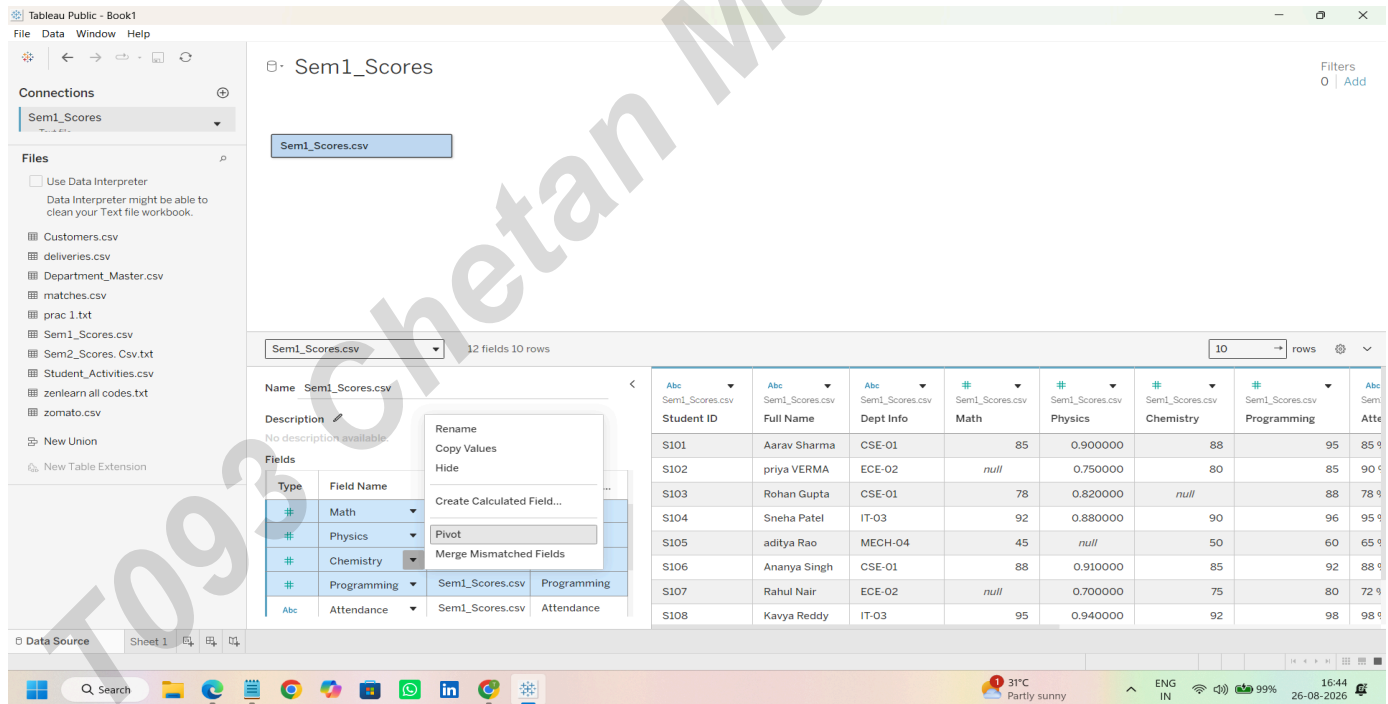


Tableau Public - Book1

File Data Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Customers.csv
deliveries.csv
Department_Master.csv
matches.csv
prac 1.txt
Sem1_Scores.csv
Sem2_Scores.Csv.txt
Student_Activities.csv
zenlearn all codes.txt
zomato.csv

New Union
New Table Extension

Sem1_Scores

Sem1_Scores.csv

12 fields 10 rows

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
#	Math	Sem1_Scores.csv	Math
#	Physics	Sem1_Scores.csv	Physics
#	Chemistry	Sem1_Scores.csv	Chemistry
#	Programming	Sem1_Scores.csv	Programming
Abc	Attendance	Sem1_Scores.csv	Attendance

Context Menu:

- Rename
- Copy Values
- Hide
- Create Calculated Field...
- Pivot
- Merge Mismatched Fields

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85%
S102	priya VERMA	ECE-02	null	0.750000	80	85	90%
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78%
S104	Sneha Patel	IT-03	92	0.880000	90	96	95%
S105	aditya Rao	MECH-04	45	null	50	60	65%
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88%
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72%
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98%

Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

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Step 3:- Double-click the header named Pivot Field Names and type: Subject Double-click the header named Pivot Field Values and type: Score

Tableau Public - Book1

File Data Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Customers.csv
deliveries.csv
Department_Master.csv
matches.csv
prac 1.txt
Sem1_Scores.csv
Sem2_Scores.Csv.txt
Student_Activities.csv
zenlearn all codes.txt
zomato.csv

New Union
New Table Extension

Sem1_Scores

Sem1_Scores.csv

10 fields 40 rows

40 rows

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Text	Attendance	Sem1_Scores.csv	Attendance
Text	Subject	Pivot	Pivot Field Na...
Text	Score	Pivot	Pivot Field Val...
Text	Physics_Cle...	Calculation	Calculation_2...
Text	Attendance_C	Calculation	Calculation_2...

Student ID	Full Name	Dept Info	Attendance	Subject	Pivot Score	Calculation
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	n
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	n
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	n
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	n
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	n
S102	priya VERMA	ECE-02	90 %	Math	null	n
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	n
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	n

Step 4:- Click the Abc icon directly above the Score column header. Select Number (decimal) from the dropdown list.

Tableau Public - Book1

File Data Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Customers.csv
deliveries.csv
Department_Master.csv
matches.csv
prac 1.txt
Sem1_Scores.csv
Sem2_Scores.Csv.txt
Student_Activities.csv
zenlearn all codes.txt
zomato.csv

New Union
New Table Extension

Sem1_Scores

Sem1_Scores.csv

10 fields 40 rows

40 rows

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Text	Attendance	Sem1_Scores.csv	Attendance
Text	Subject	Pivot	Pivot Field Na...
Text	Score	Pivot	Pivot Field Val...
Text	Physics_Cle...	Calculation	Calculation_2...
Text	Attendance_C	Calculation	Calculation_2...

Student ID	Full Name	Dept Info	Attendance	Subject	Pivot Score	Calculation
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	n
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	n
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	n
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	n
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	n
S102	priya VERMA	ECE-02	90 %	Math	null	n
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	n
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	n

Name:- Chetan Mandavkar

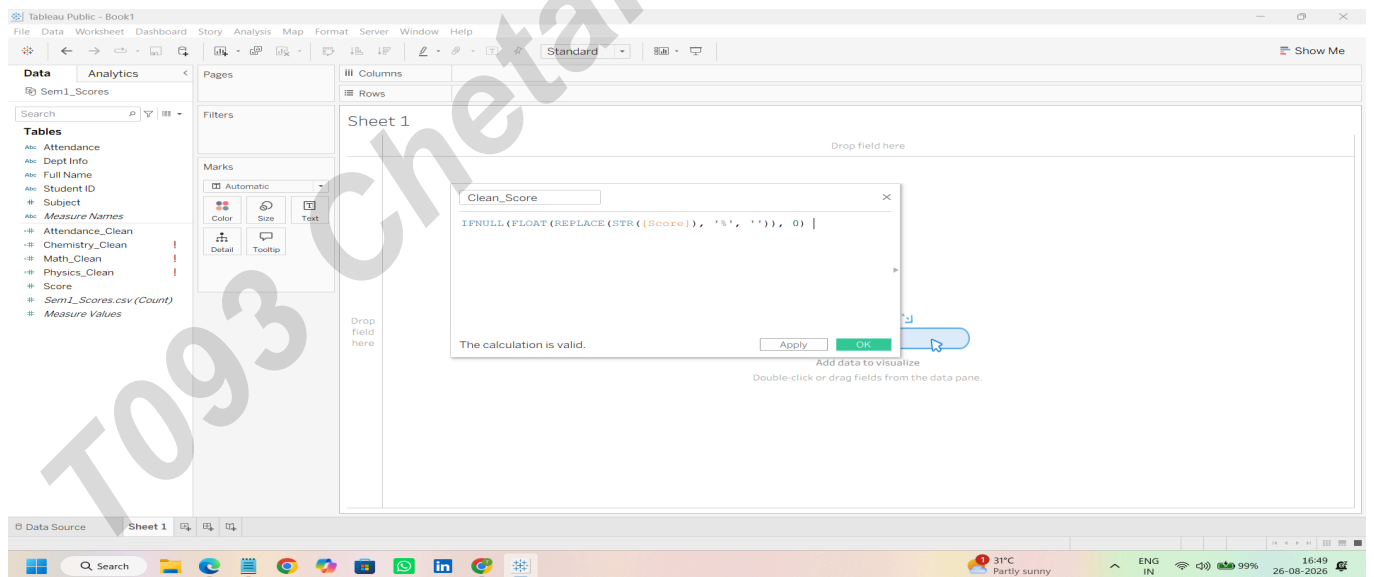
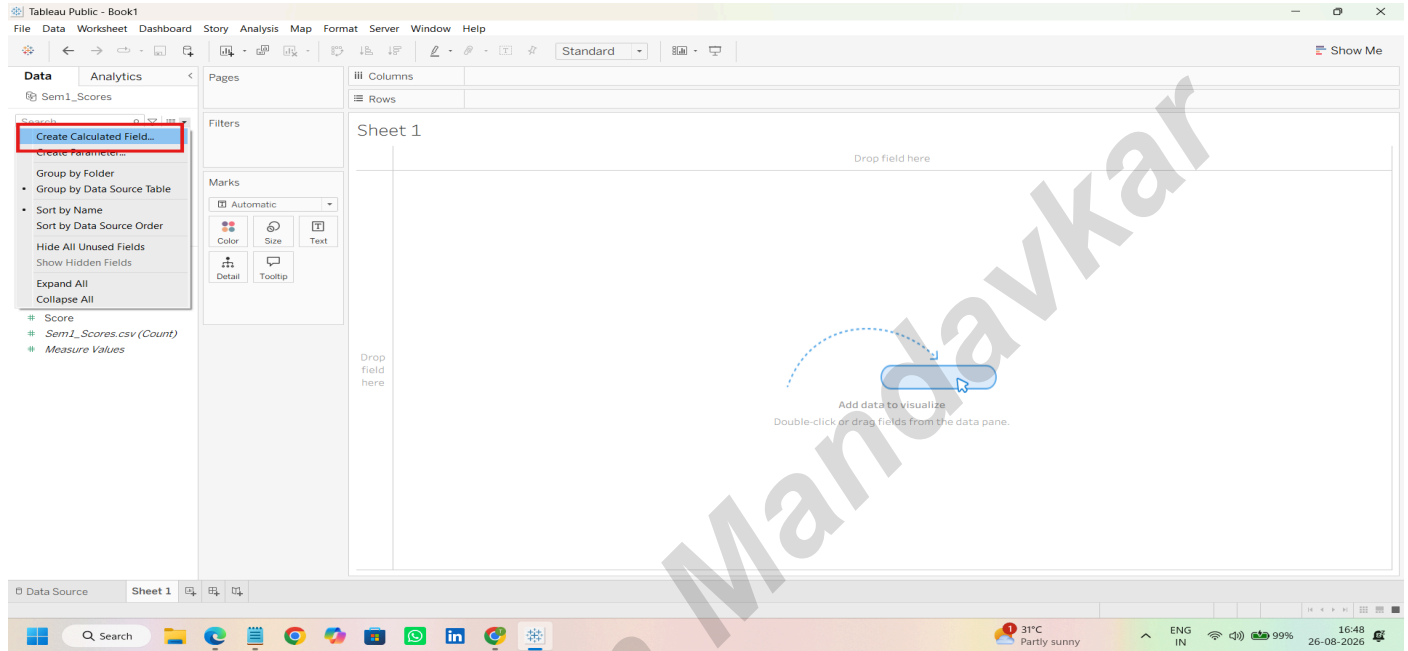
Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

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Step 5:- Alternatively, create a calculated field inside Sheet 1 named Clean_Score:

Code snippet IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0) And Click OK



Name:- Chetan Mandavkar

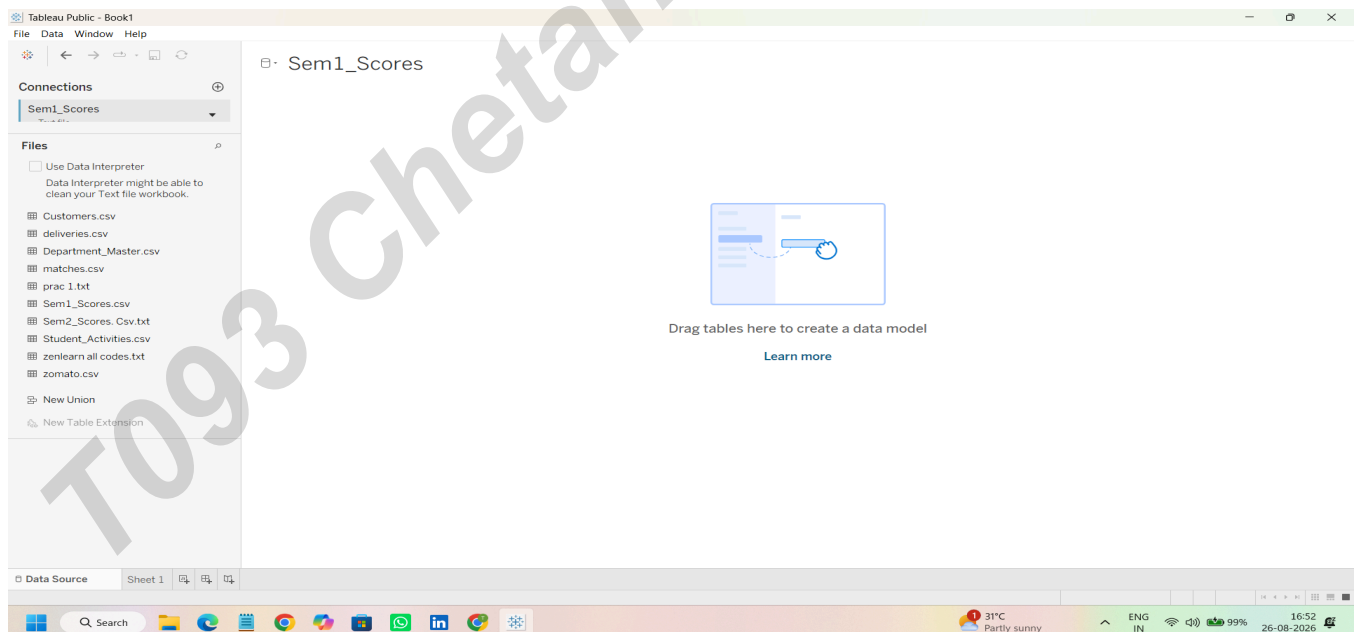
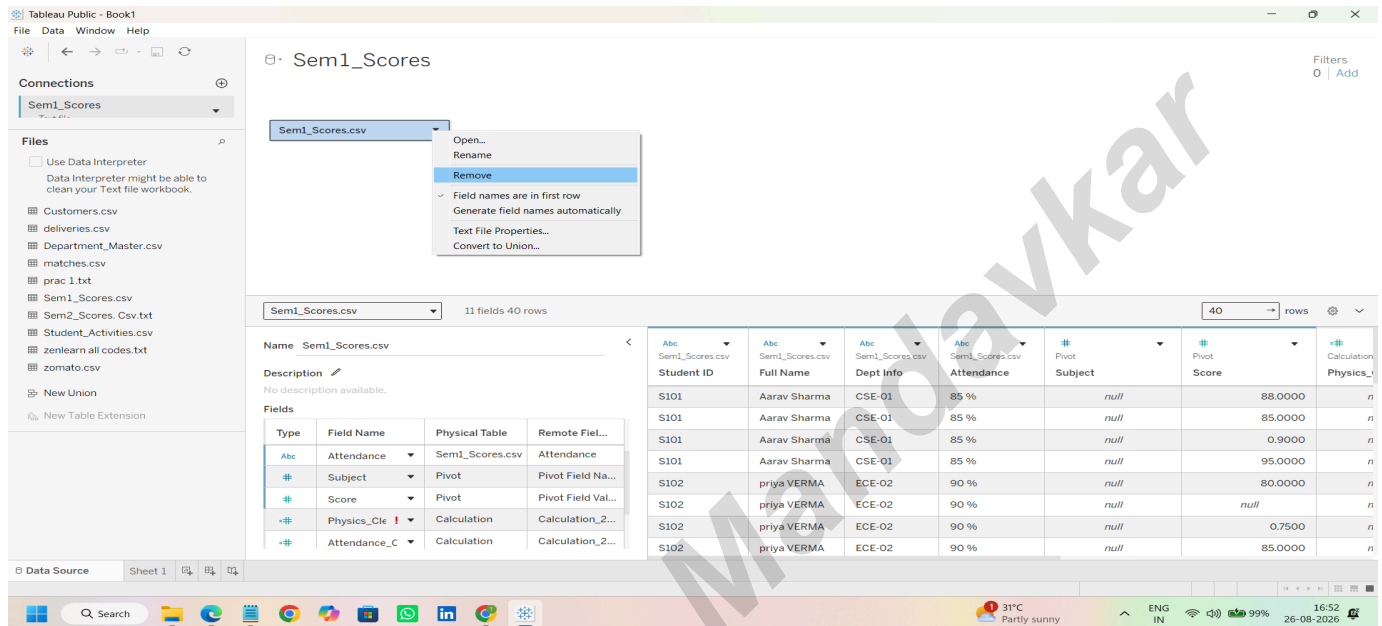
Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

Sheth L.U.J. And Sir M.V. College

C: Perform Union of Multiple Files

Step 1:- Open Tableau Desktop and go to the Data Source tab in the bottom-left corner. If any single table is already on the canvas, right-click it and select Remove to start with a clean canvas



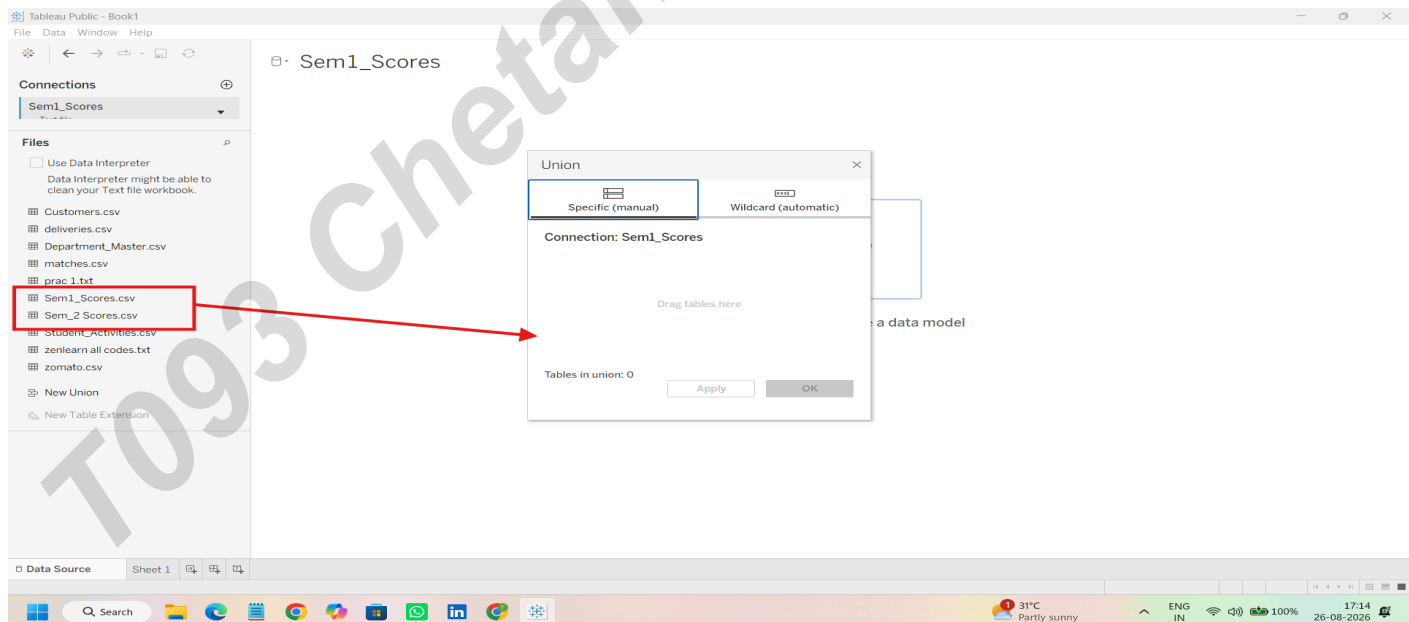
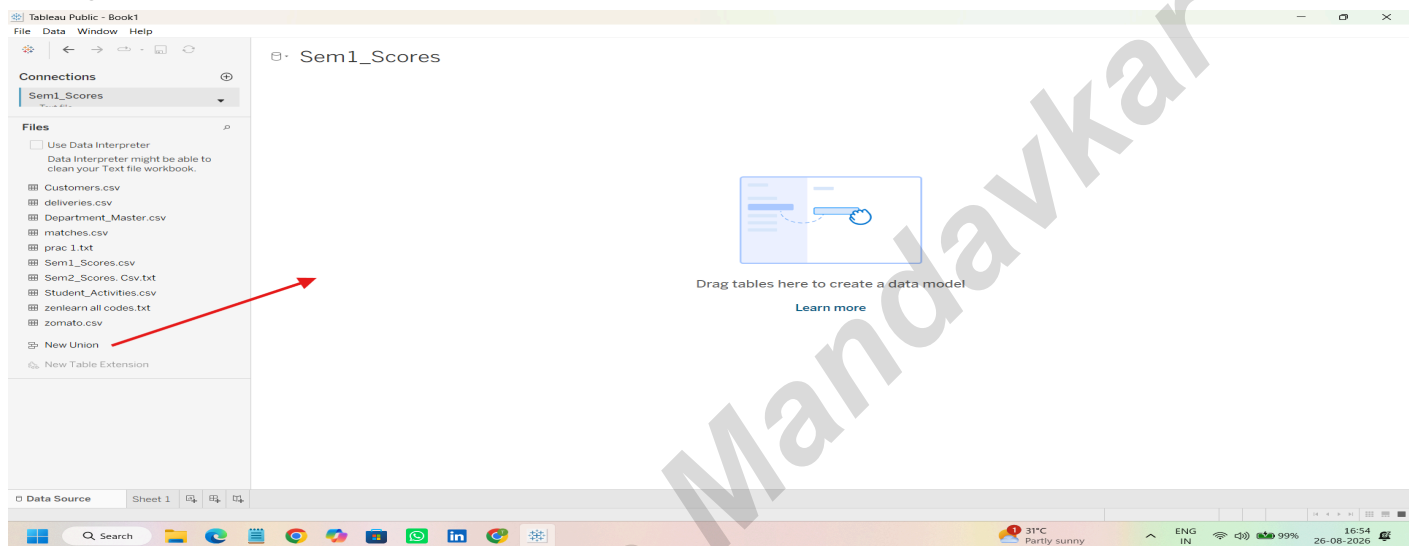
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

Sheth L.U.J. And Sir M.V. College

Step 2:- Look at the left sidebar under the Files / Connections pane. Drag the New Union option onto the empty canvas area. In the pop-up Union dialog box: o Drag Sem1_Scores.csv from the left panel into the box. o Drag Sem2_Scores.csv from the left panel into the same box. o Verify that both filenames are listed in the Union box. Click Apply, then click OK.

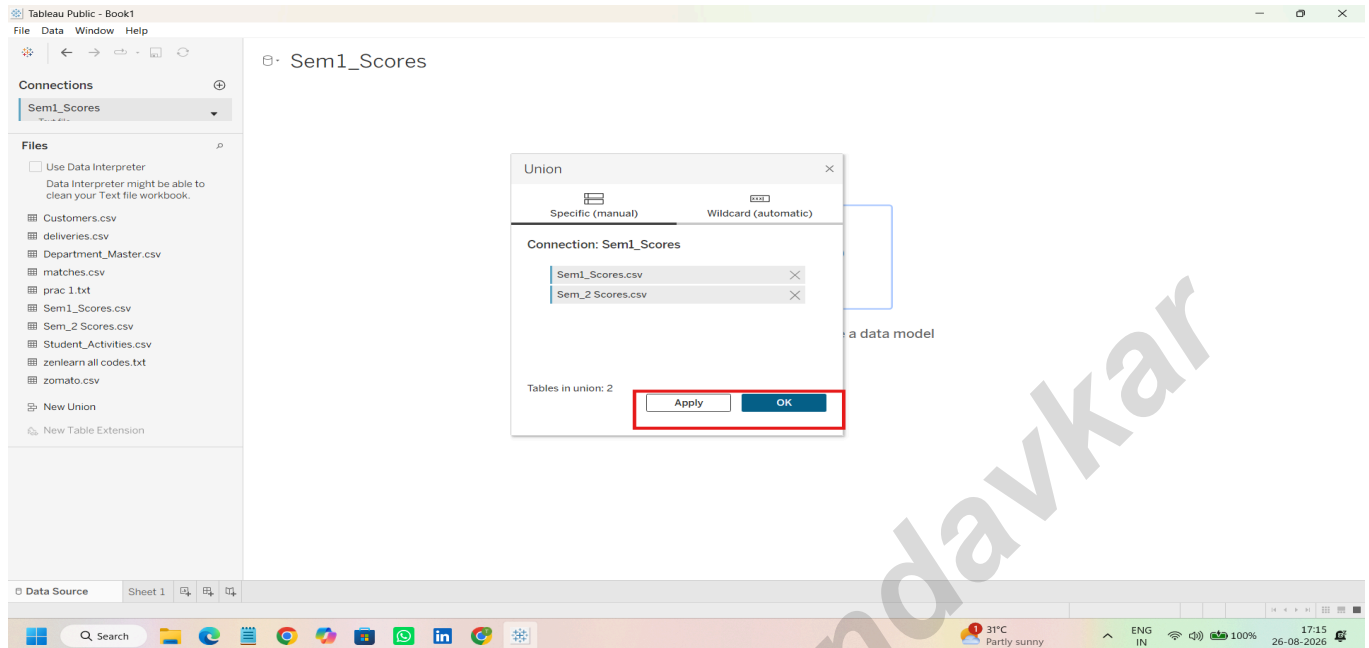


Name:- Chetan Mandavkar

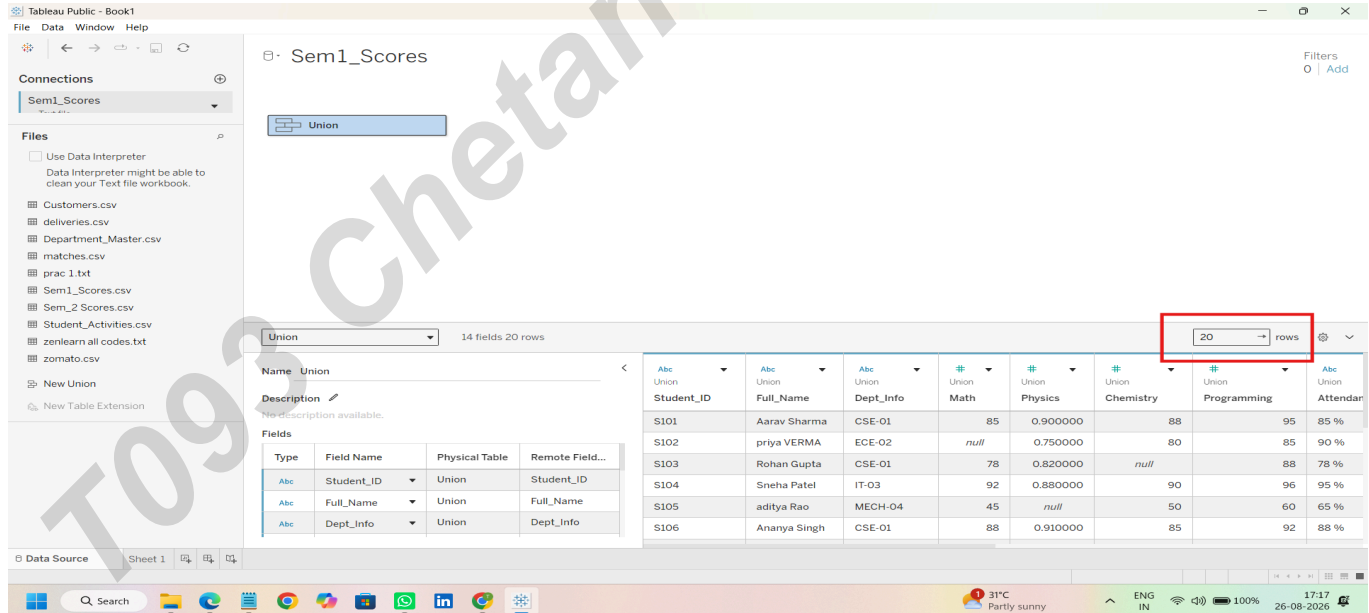
Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

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Step 3:- Review the data grid at the bottom of the screen. Confirm the row count: the total number of rows increases from 10 to 20 rows (10 records for Semester 1 and 10 records for Semester 2).



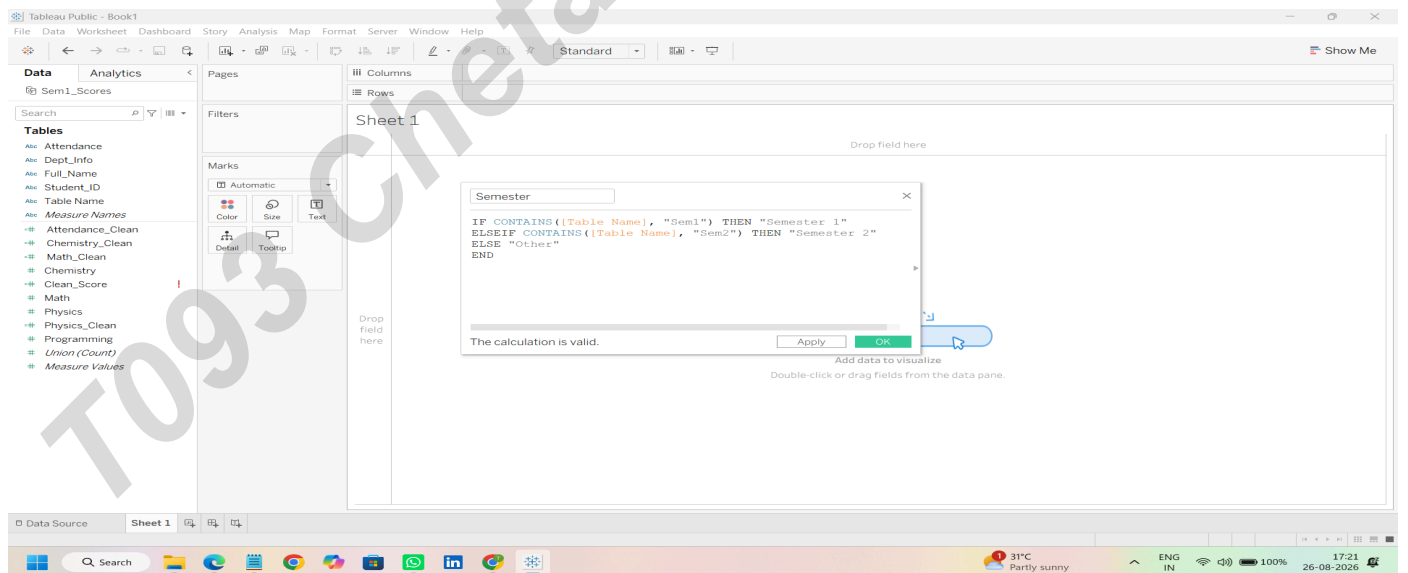
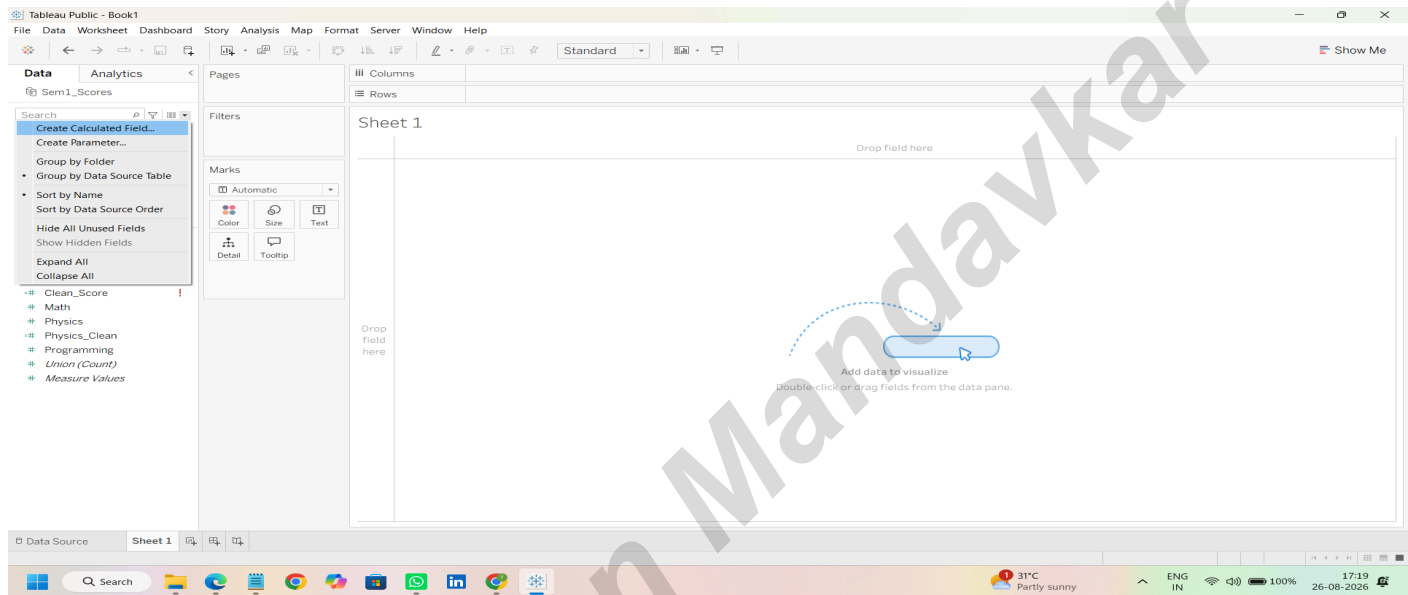
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

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Step 4:- Click on Sheet 1 at the bottom-left. 2. In the Data Pane (left sidebar), right-click in an empty space select Create Calculated Field.... 3. Set the Field Name to: Semester 4. Enter the following formula: Code snippet IF CONTAINS([Table Name], "Sem1") THEN "Semester 1" ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2" ELSE "Other" END



Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

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D: Execute Cross-Database Joins

Step 1:- In the left sidebar under Files / Connections, find Department_Master.csv. Drag Department_Master.csv onto the canvas and drop it to the right of your Union box. A flexible line (relationship / join noodle) will connect the two boxes. If the relationship dialog opens (or click the relationship line):

- o Left Table Field (Union): Select Dept_Info
- o Right Table Field (Department_Master): Select Dept_Code
- o Operator: =

Tableau Public - Book1

File Data Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Customers.csv
deliveries.csv
Department_Master.csv
matches.csv
prac 1.txt
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv
zenlearn all codes.txt
zomato.csv

New Union
New Table Extension

Sem1_Scores

Union

15 fields 20 rows

Name Union

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name
Abc	Dept_Info	Union	Dept_Info

Abc Union Student_ID	Abc Union Full_Name	Abc Union Dept_Info	Union Math	Union Physics	Union Chemistry	Union Programming	Abc Union Attendar
S101	Agrav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %

Tableau Public - Book1

File Data Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Customers.csv
deliveries.csv
Department_Master.csv
matches.csv
prac 1.txt
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv
zenlearn all codes.txt
zomato.csv

New Union
New Table Extension

Sem1_Scores

Union

Department_Master.csv

Union — Departme...

Name Department_Master.csv

How do relationships differ from joins? Learn more

Union	Operator	Department_Maste...
Abc Dept_Info	=	Abc Dept Code

Add more fields

Performance Options

Remove Relationship

Abc Department_Master.csv Dept Code	Abc Department_Master.csv Department Name	Abc Department_Master.csv HOD Name	Department_Master.csv Dept Target Average	Department_Master.csv Total Dept Capacity
CSE-01	Computer Science	Dr. Raman	85	120
ECE-02	Electronics & Comm	Dr. Iyer	78	90
IT-03	Information Tech	Dr. Sen	82	60
MECH-04	Mechanical Engg	Dr. Sharma	70	60

Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

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Step 2:- From the left sidebar, drag Student_Activities.csv onto the canvas. Drop it to the right of the Union box. In the relationship dialog:

- o Left Table Field (Union): Select Student_ID
- o Right Table Field (Student_Activities): Select Student_ID
- o Operator: =

The screenshot shows the Tableau Public interface with a workbook named 'Sem1_Scores'. The 'Connections' pane on the left lists several files, including 'Student_Activities.csv'. The 'Files' pane shows a list of files, with 'Student_Activities.csv' highlighted. The main canvas displays a Union relationship between 'Department_Master.csv' and 'Student_Activities.csv'. The relationship dialog is open, showing the 'Name' as 'Student_Activities.csv' and the 'How do relationships differ from joins?' section. The 'Union' section shows 'Student_ID' from 'Student_Activities.csv' selected for both the left and right tables, with the operator set to '='. The 'Performance Options' section is also visible. The data preview on the right shows a table with columns: 'Student ID (Student Acti...', 'Club Membership', 'Projects Completed', and 'Certifications'. The table contains 6 rows of data.

This screenshot is similar to the one above, but with a red box highlighting the relationship dialog. The dialog shows the 'Name' as 'Student_Activities.csv' and the 'How do relationships differ from joins?' section. The 'Union' section shows 'Student_ID' from 'Student_Activities.csv' selected for both the left and right tables, with the operator set to '='. The 'Performance Options' section is also visible. The data preview on the right shows a table with columns: 'Student ID (Student Acti...', 'Club Membership', 'Projects Completed', and 'Certifications'. The table contains 6 rows of data.

Name:- Chetan Mandavkar

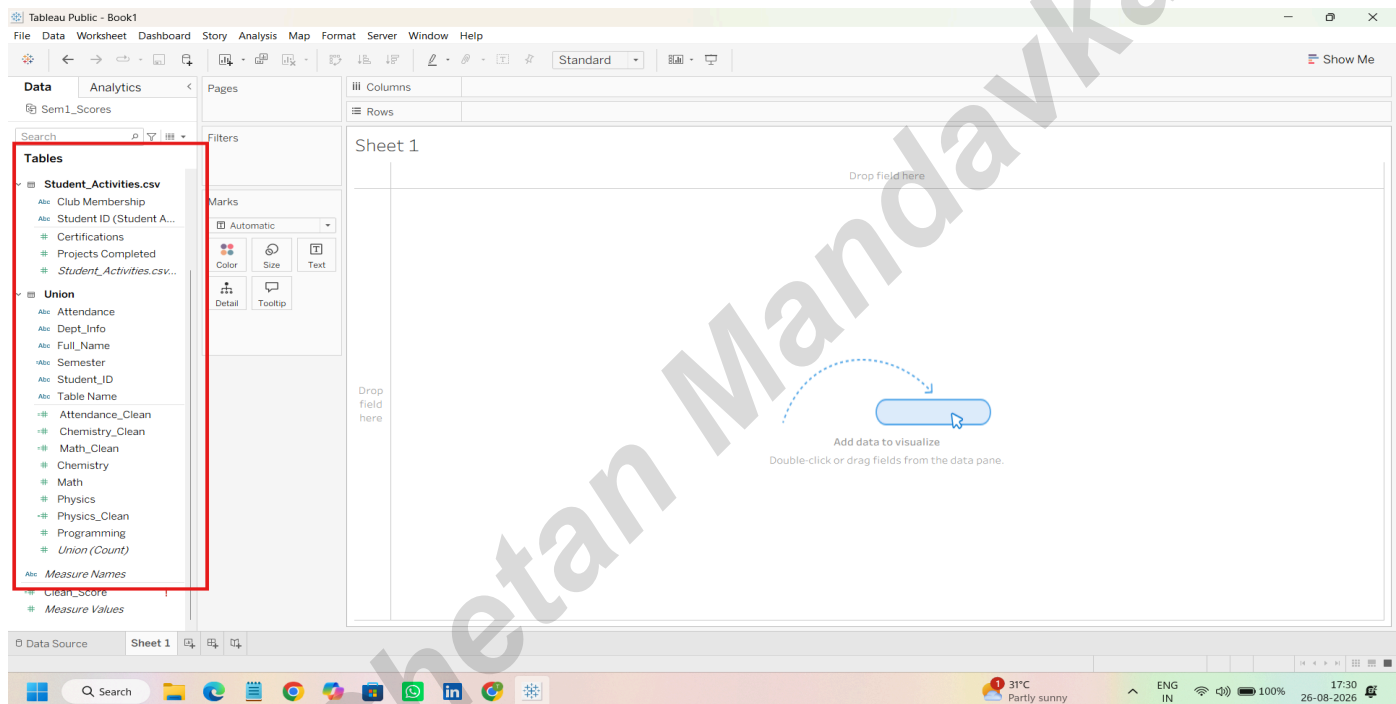
Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

Sheth L.U.J. And Sir M.V. College

Step 3:- Click on Sheet 1 at the bottom-left. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:

- o Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
- o Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
- o Student_Activities: Club_Membership, Projects_Completed, Certifications



Name:- Chetan Mandavkar

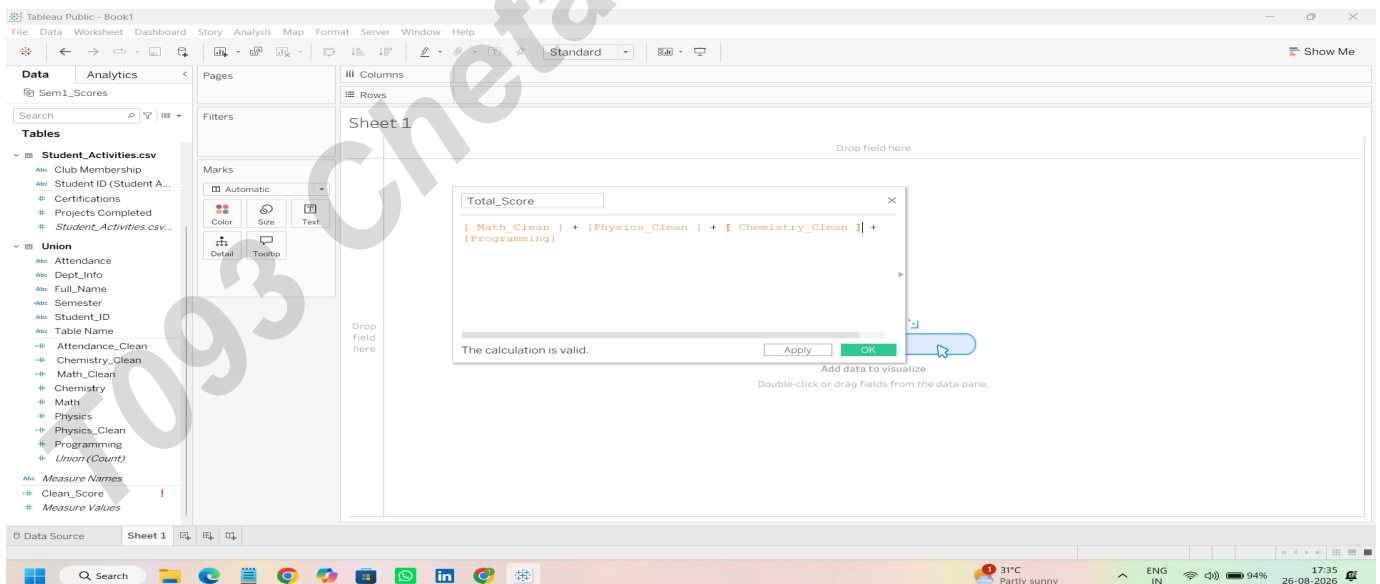
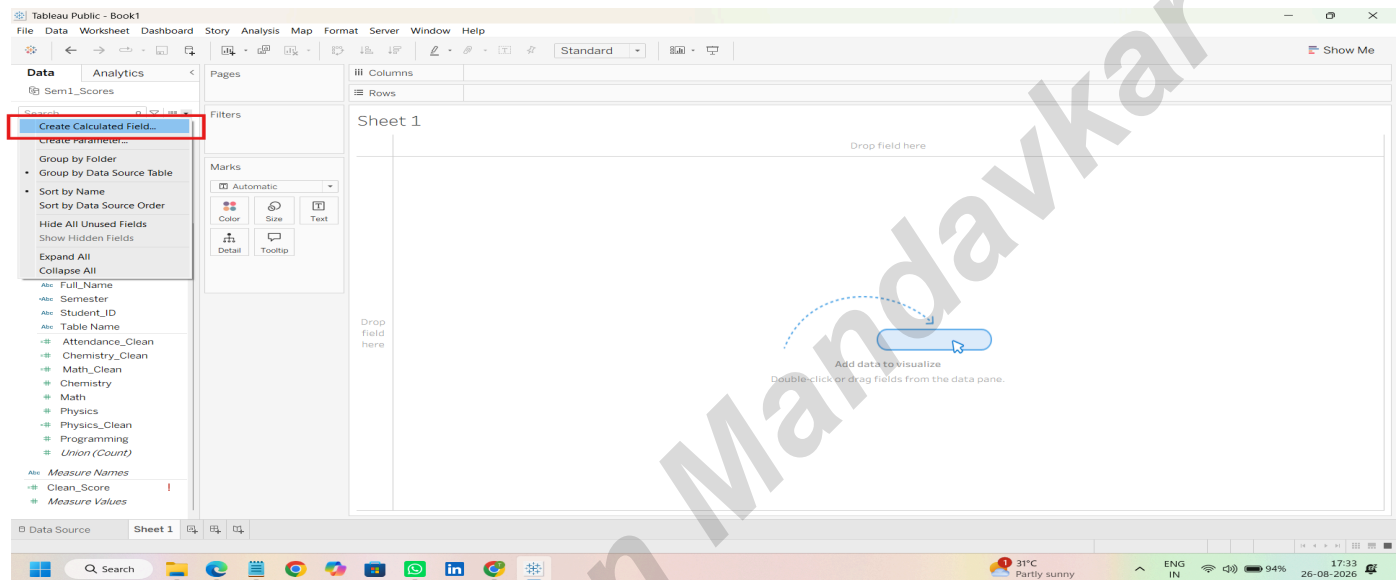
Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

Sheth L.U.J. And Sir M.V. College

E: Restructure Data for Better Visualization

Step 1:- Navigate to Sheet 1 in Tableau. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field.... Create the student's total marks field: o Field Name: Total_Score o Tableau Formula: Code snippet [Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming] o Click OK.



Name:- Chetan Mandavkar

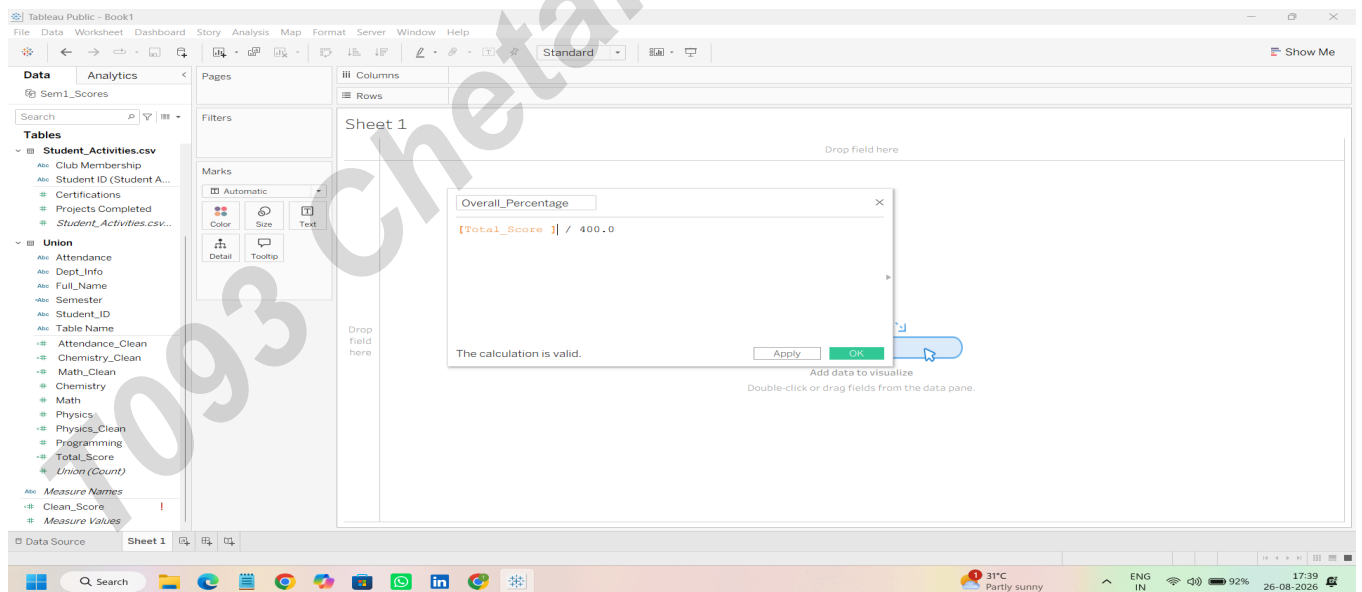
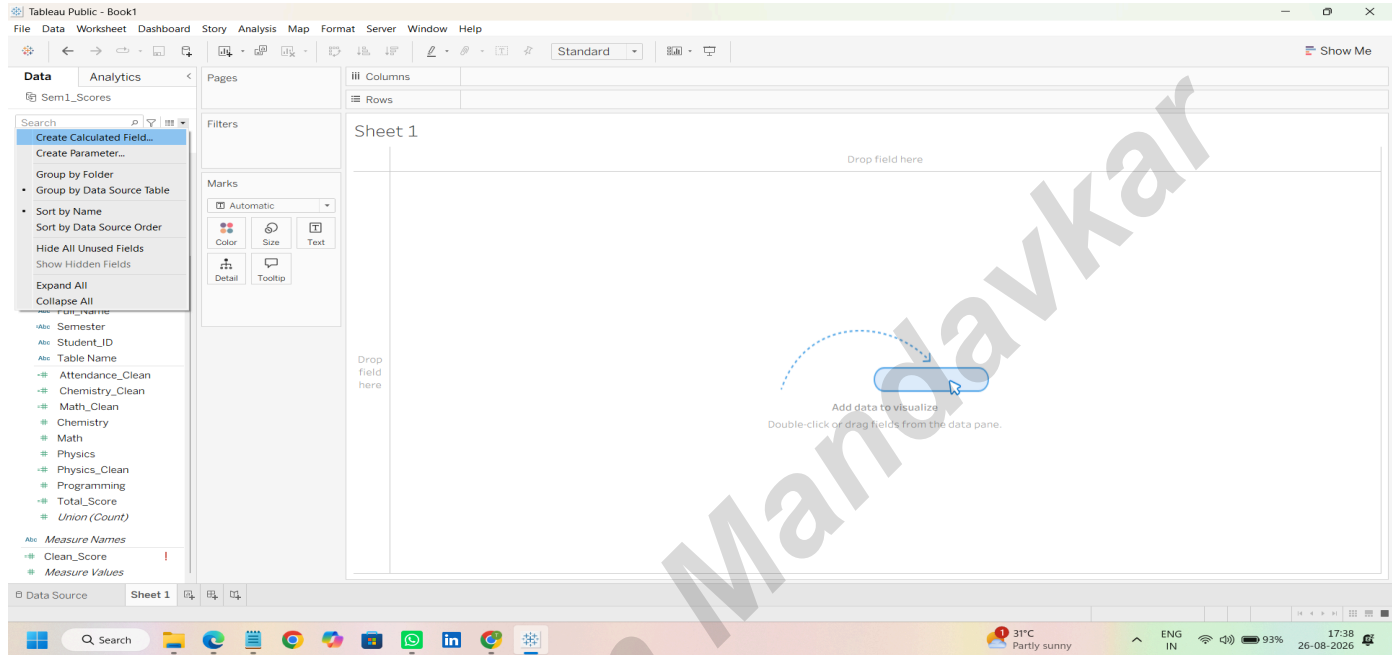
Roll No. T093

Subject:- Data Visualization With Power BI And Tableau

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Step 2:- Create the scaled overall percentage field:

- o Right-click in the Data Pane --> select **Create Calculated Field....**
- o Field Name: **Overall_Percentage**
- o Tableau Formula: Code snippet **[Total_Score] / 400.0**
- o Click **OK**.



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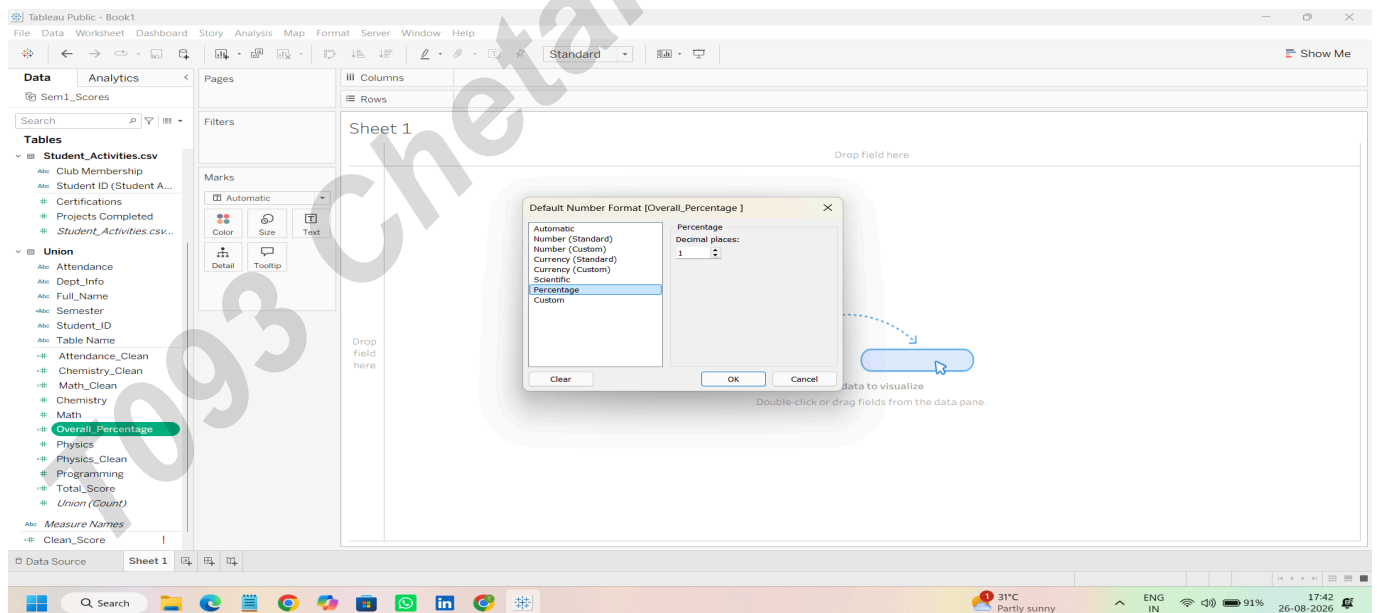
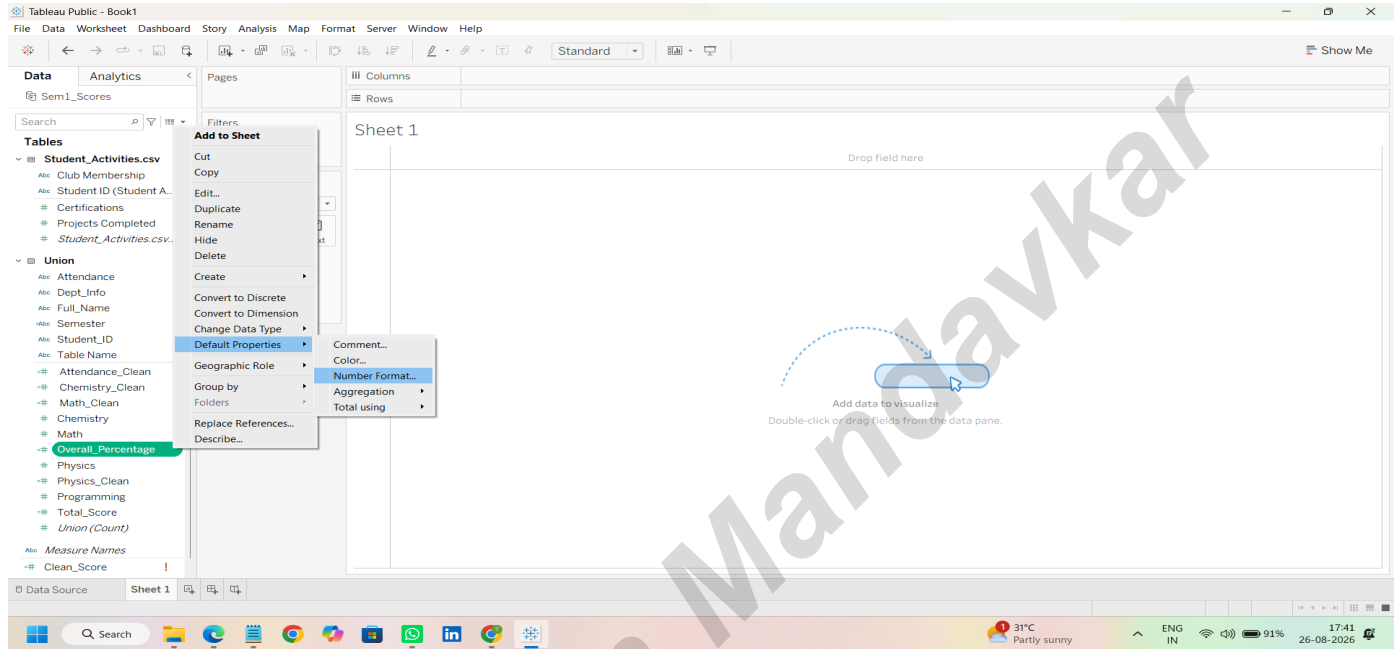
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Step 3:- Set the default percentage display formatting:

- o In the Data Pane, right-click Overall_Percentage -->Default Properties - >Number Format....
- Select Percentage and set decimal places to 1 (e.g., 85.5%).
- o Click OK.



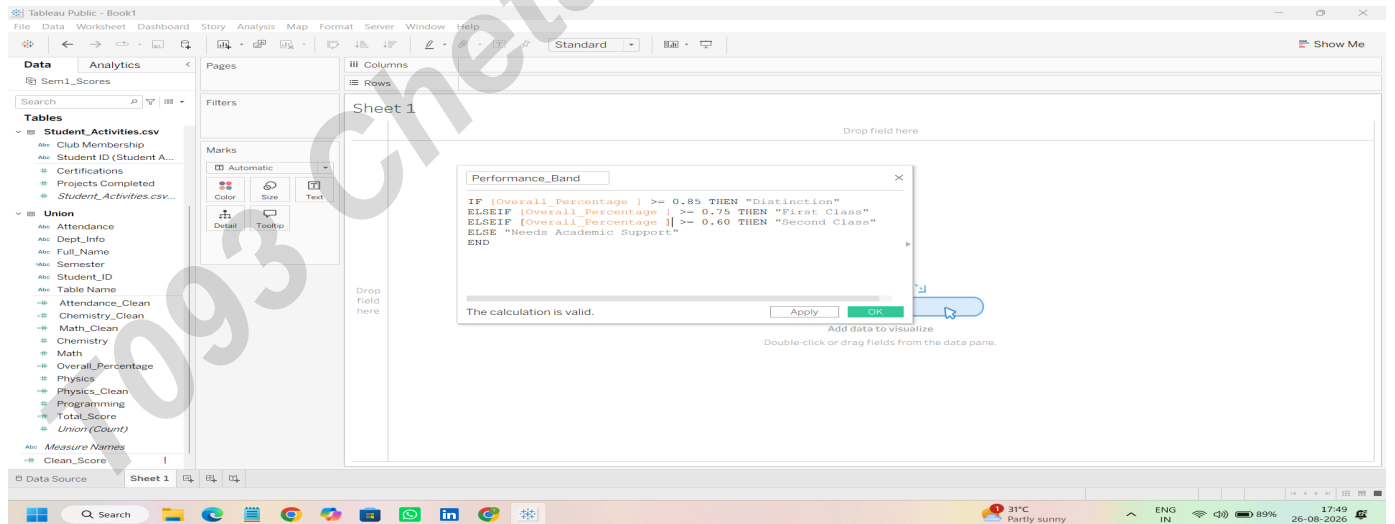
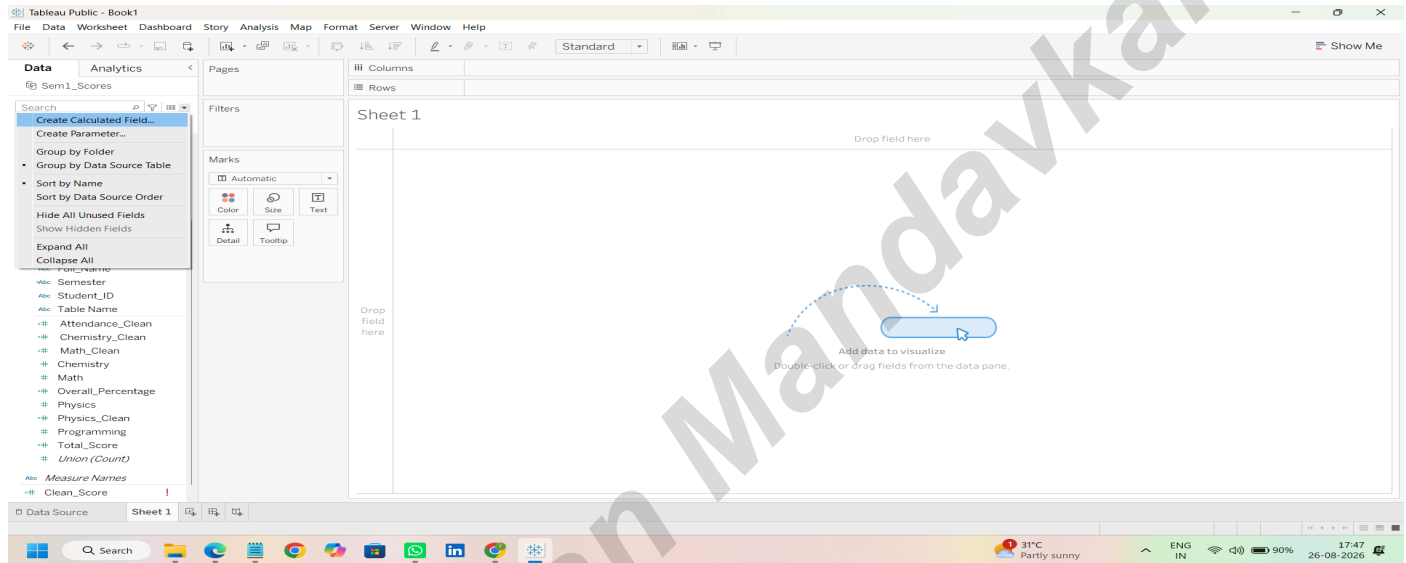
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Step 4:- Right-click in the Data Pane -->select Create Calculated Field.... Create the academic grading tier field: o Field Name: Performance_Band o Tableau Formula: Code snippet IF [Overall_Percentage] >= 0.85 THEN "Distinction" ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class" ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class" ELSE "Needs Academic Support" END o Click OK.



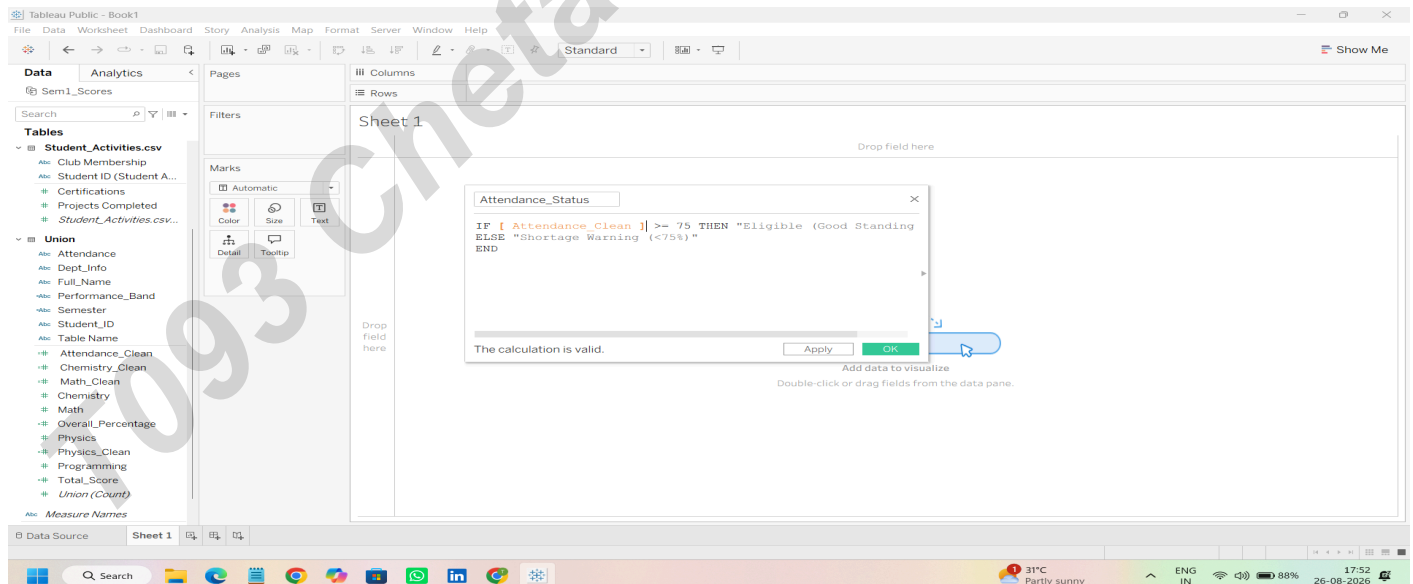
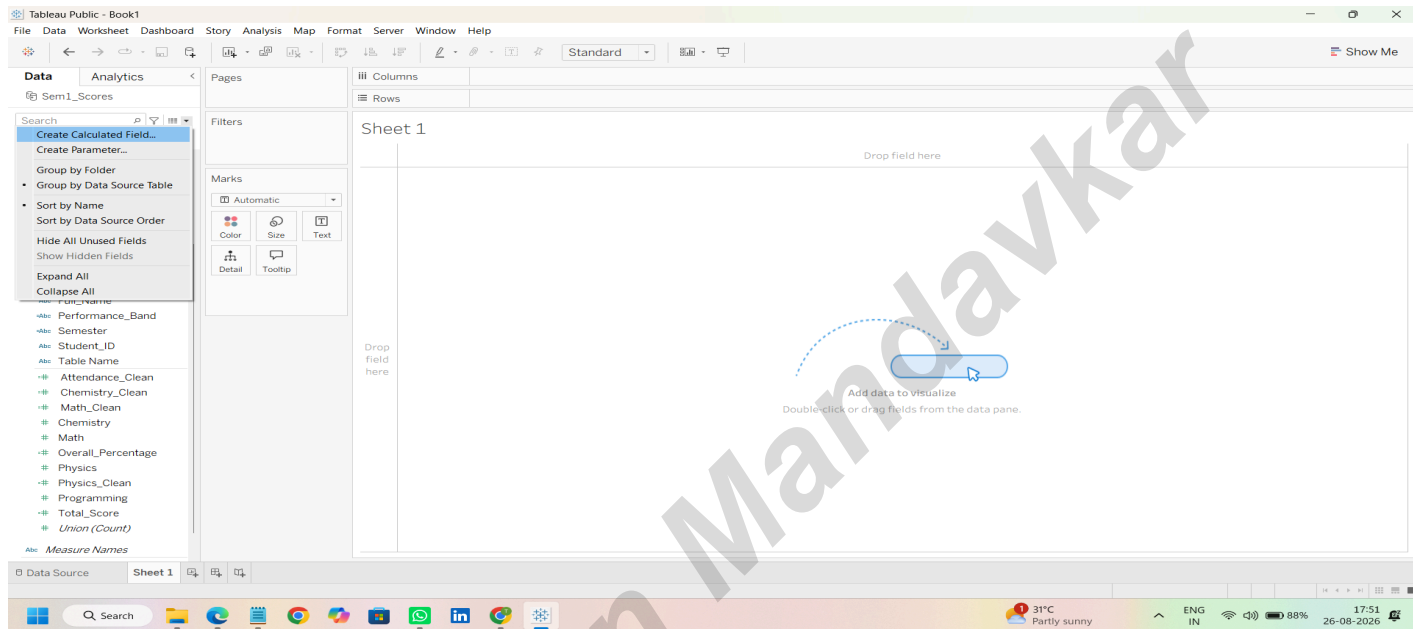
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Step 5:- Right-click in the Data Pane -->select Create Calculated Field....
2. Create the attendance compliance indicator:
o **Field Name:** Attendance_Status
o **Tableau Formula:** Code snippet IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END
o **Click OK.**



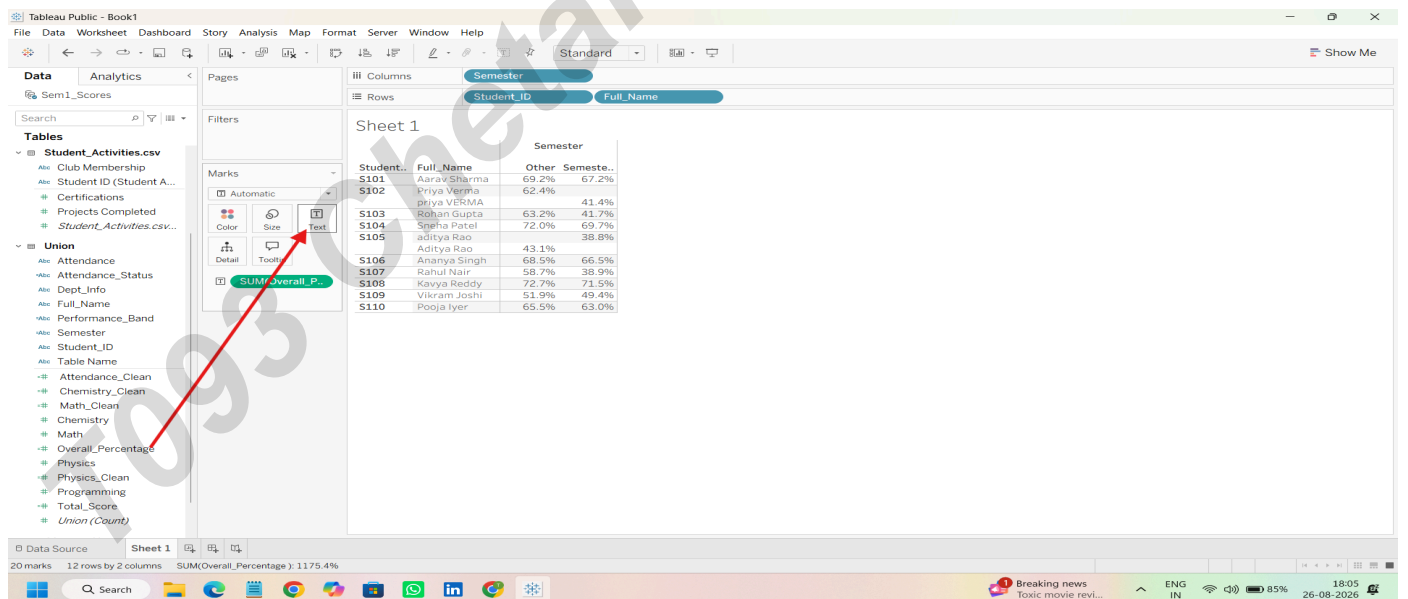
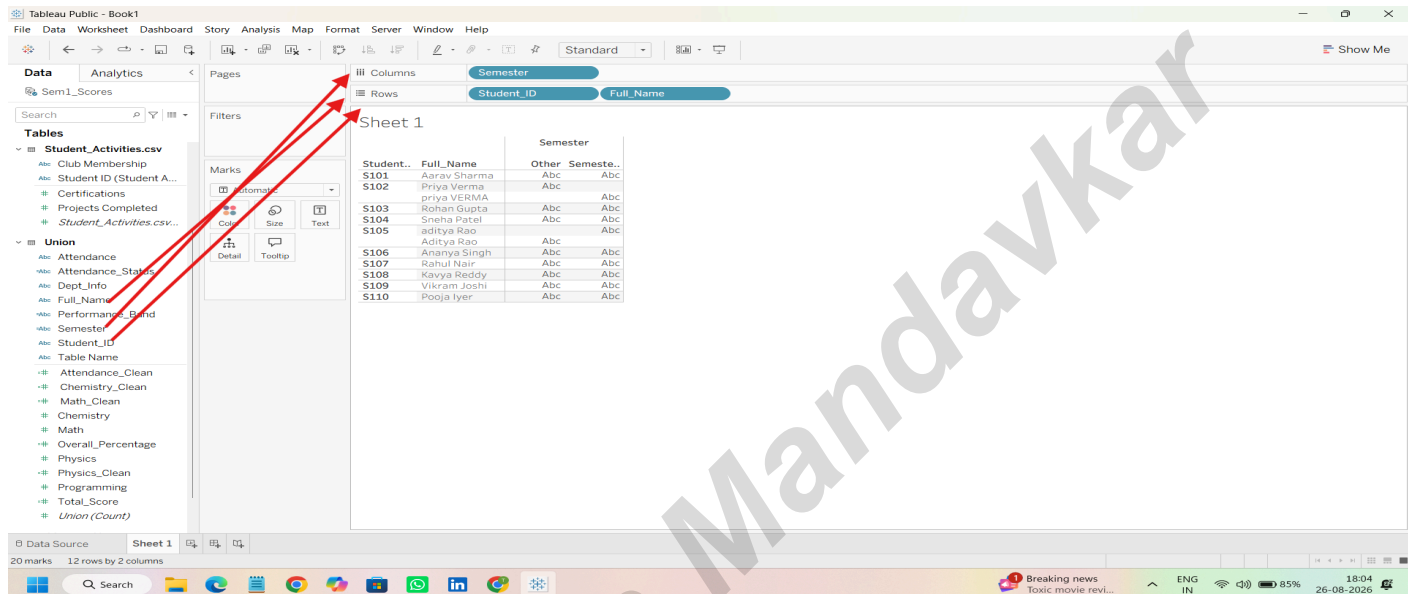
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Step 6:- Drag Student_ID and Full_Name onto the Rows shelf. Drag Semester onto the Columns shelf. Drag Overall_Percentage onto the Text tile on the Marks card. Drag Performance_Band onto the Color tile on the Marks card. Drag Attendance_Status onto the Tooltip (or Detail) tile on the Marks card.

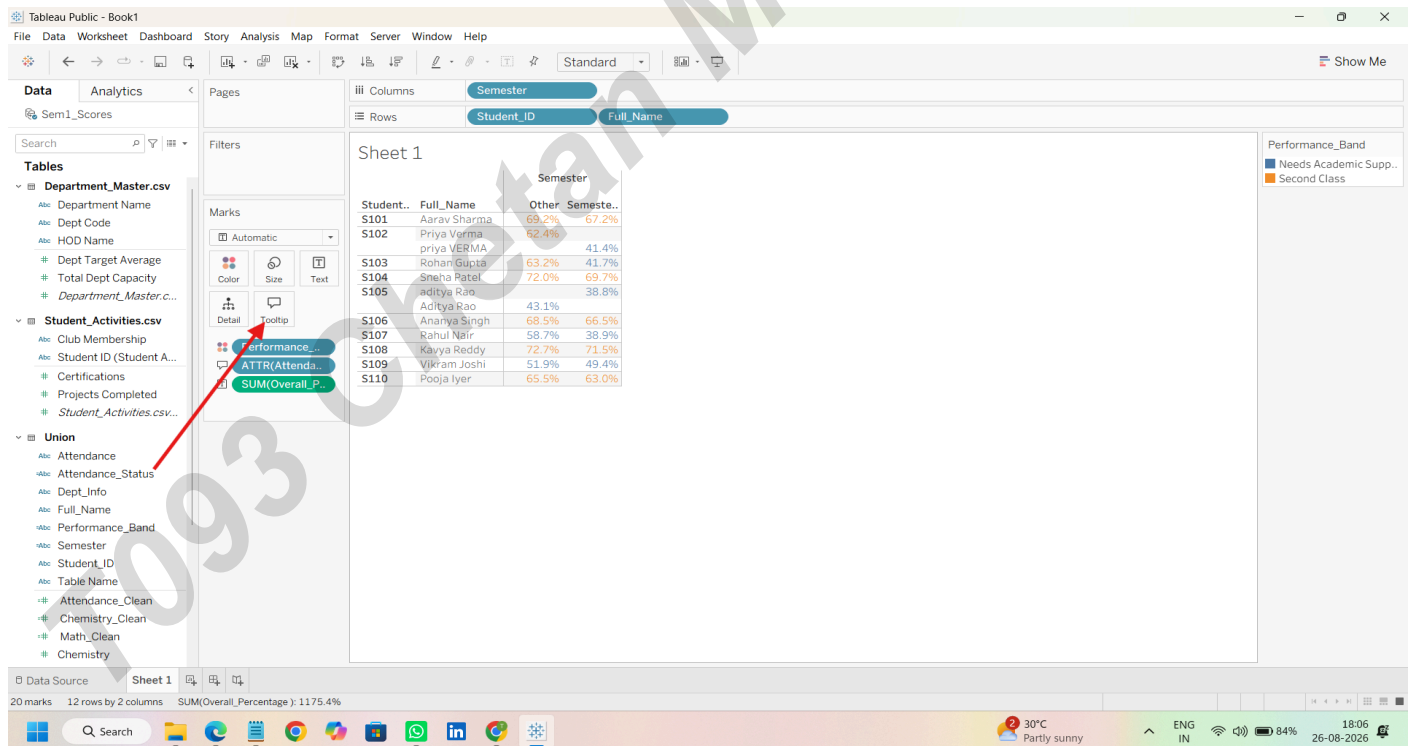
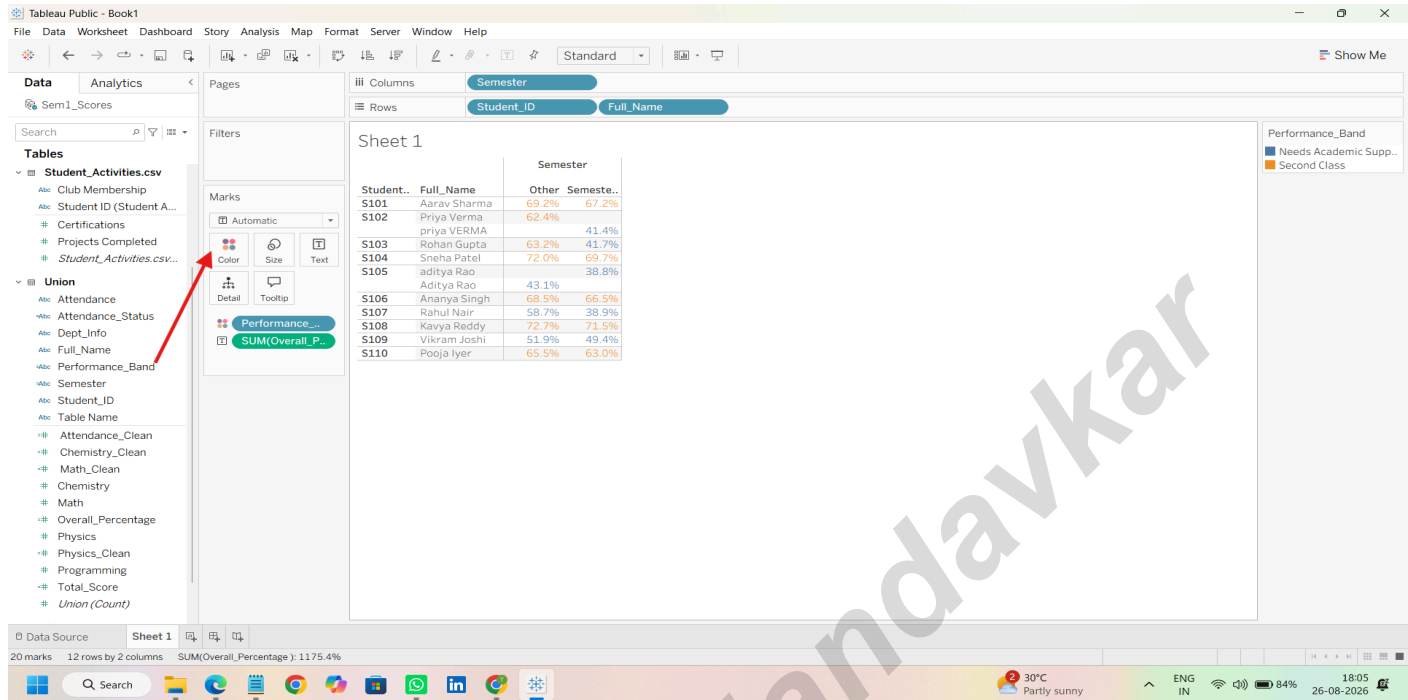


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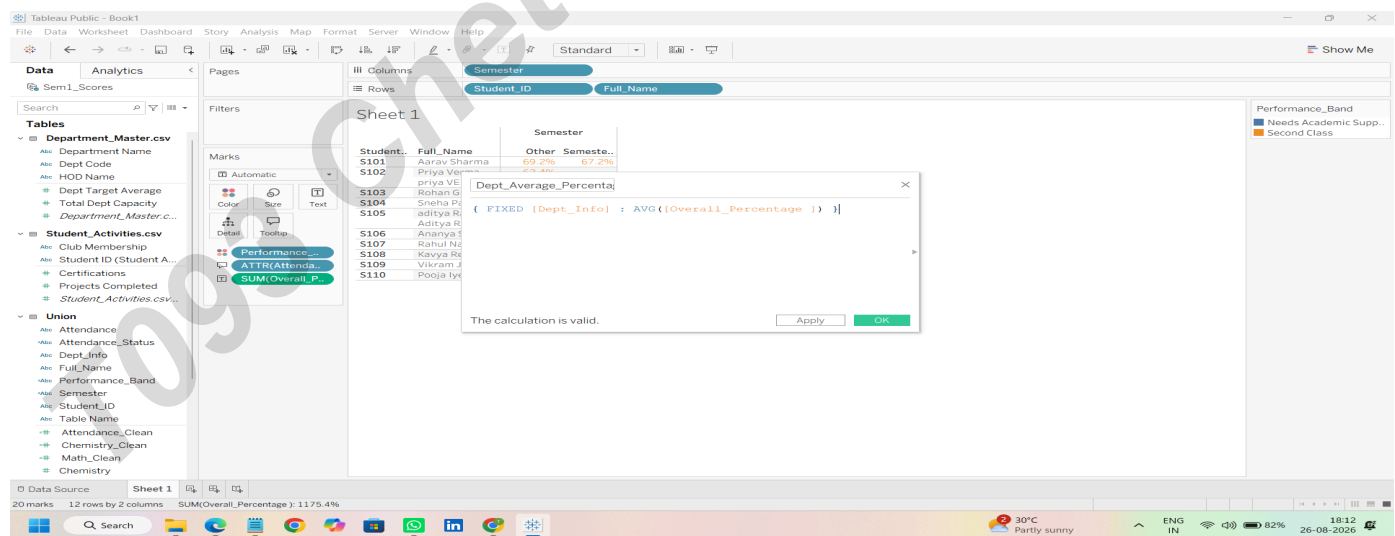
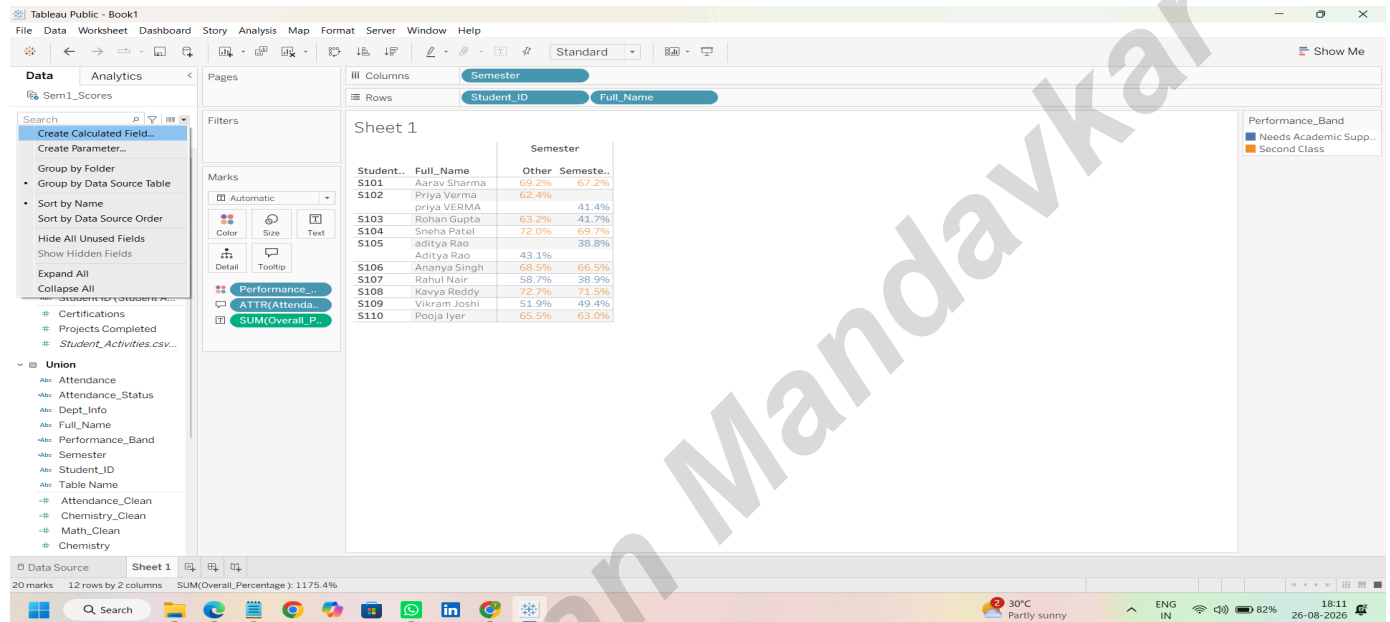
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F: Handle Different Levels of Detail in Datasets

Step 1:- In the Data Pane on the left, right-click an empty space --> select **Create Calculated Field...** Create the department-wide benchmark field: o Field Name: **Dept_Average_Percentage_LOD** o Tableau Formula: Code snippet { FIXED [Dept_Info] : AVG([Overall_Percentage]) } o Click OK



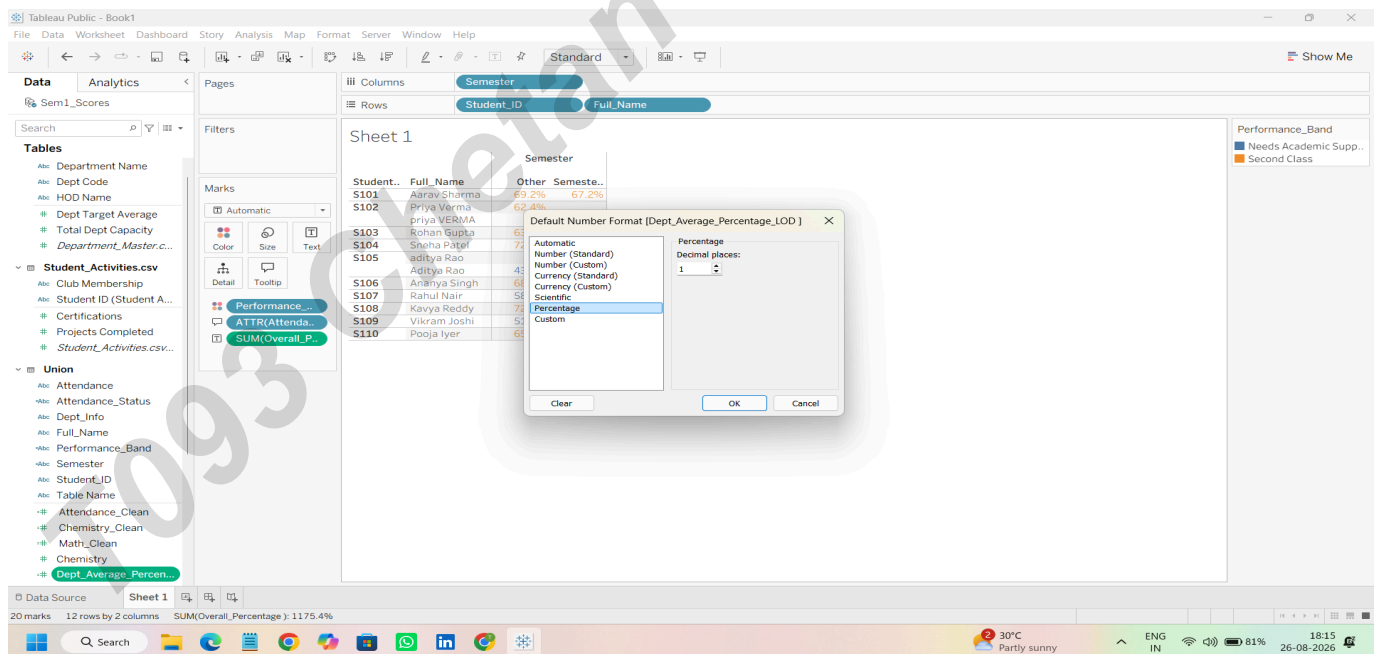
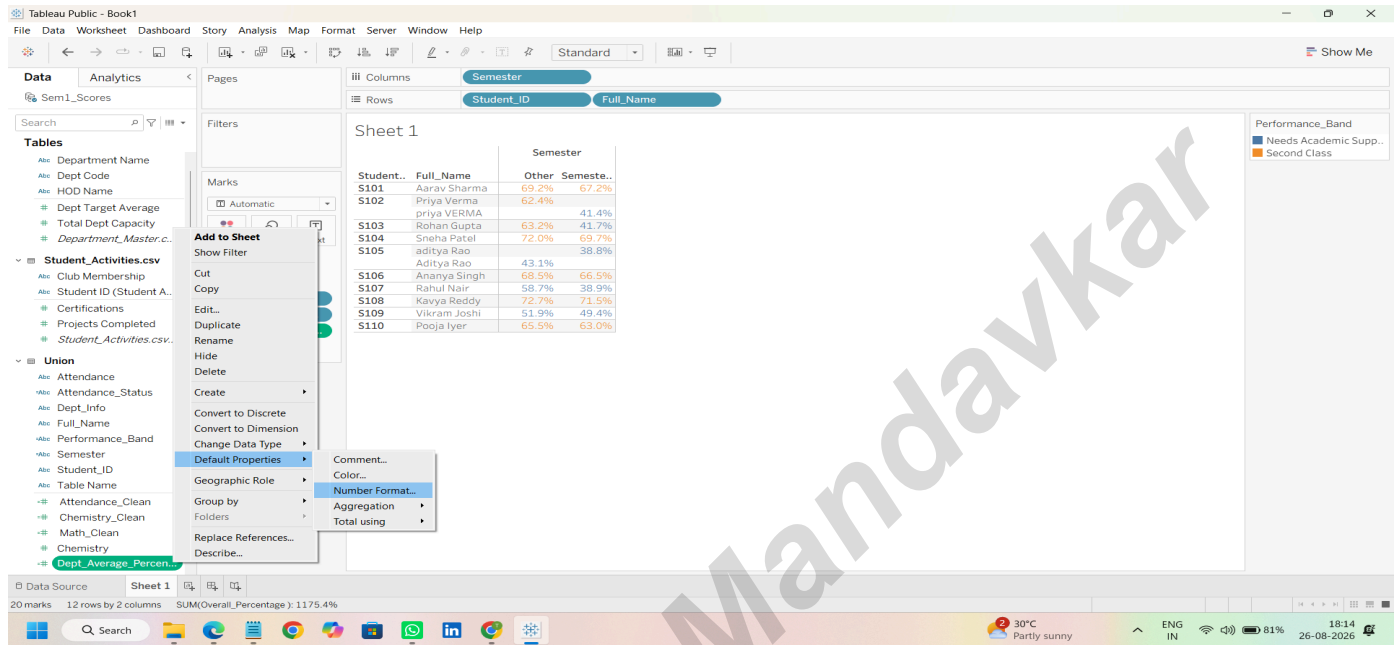
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**Step 2:- Set default formatting: o Right-click
Dept_Average_Percentage_LOD --> Default Properties --> Number
Format... --> select Percentage (1 decimal place) --> OK.**



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Step 3:- Right-click in the Data Pane --> select Create Calculated Field.... Create the cumulative student benchmark across all semesters: o Field Name: Student_Overall_Career_Average o Tableau Formula: Code snippet { FIXED [Student_ID] : AVG([Overall_Percentage]) } o Click OK.

Student_ID	Full_Name	Semester	Overall_Percentage
S101	Aarav Sharma	Other	69.2%
S102	Priya Verma	Other	67.2%
S103	Rohan Gupta	Other	63.2%
S104	Sneha Patel	Other	72.0%
S105	Aditya Rao	Other	38.8%
S106	Ananya Singh	Other	68.5%
S107	Rahul Nair	Other	58.7%
S108	Kavya Reddy	Other	72.7%
S109	Vikram Joshi	Other	51.9%
S110	Pooja Iyer	Other	65.5%

Student_ID	Full_Name	Semester	Overall_Percentage
S101	Aarav Sharma	Other	69.2%
S102	Priya Verma	Other	67.2%
S103	Rohan Gupta	Other	63.2%
S104	Sneha Patel	Other	72.0%
S105	Aditya Rao	Other	38.8%
S106	Ananya Singh	Other	68.5%
S107	Rahul Nair	Other	58.7%
S108	Kavya Reddy	Other	72.7%
S109	Vikram Joshi	Other	51.9%
S110	Pooja Iyer	Other	65.5%

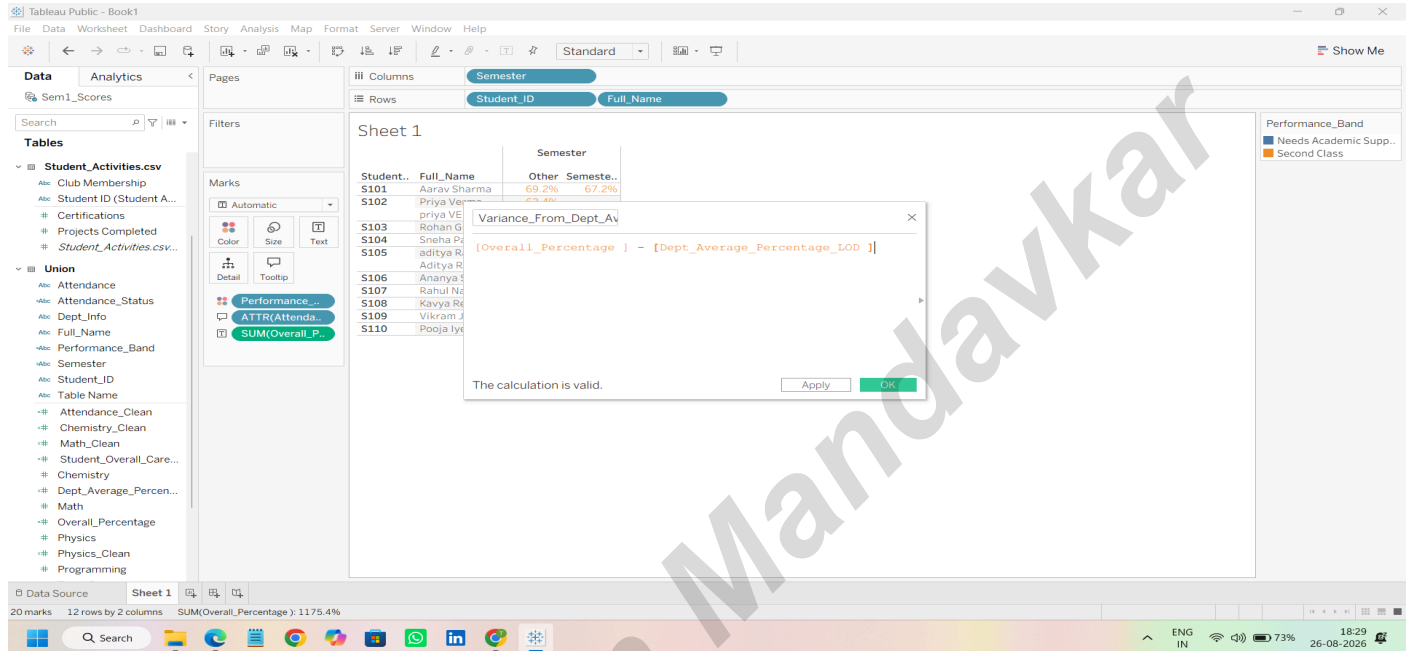
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Step 4:-Right-click in the Data Pane --> select Create Calculated Field.... 2. Create the relative difference metric: o Field Name: Variance_From_Dept_Average o Tableau Formula: Code snippet [Overall_Percentage] - [Dept_Average_Percentage_LOD] o Click OK.



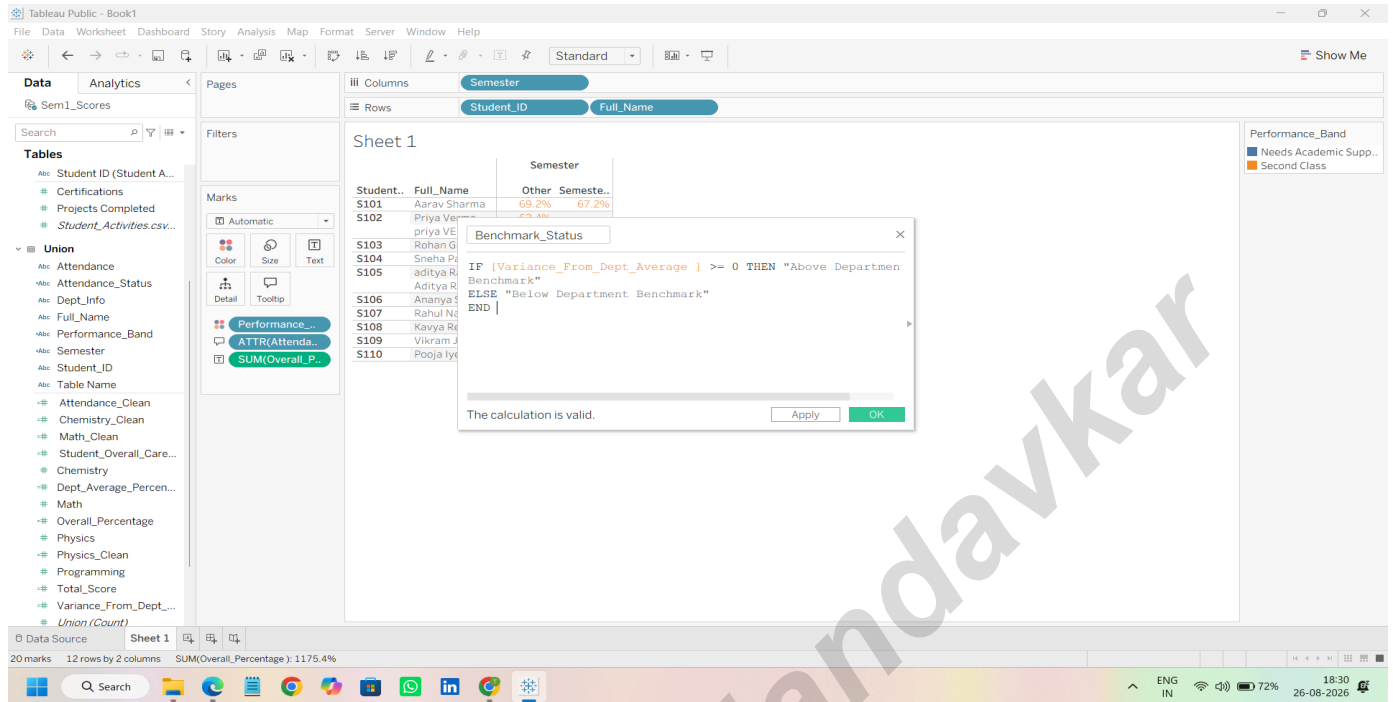
Step 5:- Right-click in the Data Pane -->select Create Calculated Field.... 2. Create an automated comparison label: o Field Name: Benchmark_Status o Tableau Formula: Code snippet IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark" ELSE "Below Department Benchmark" END o Click OK.

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Step 6:-

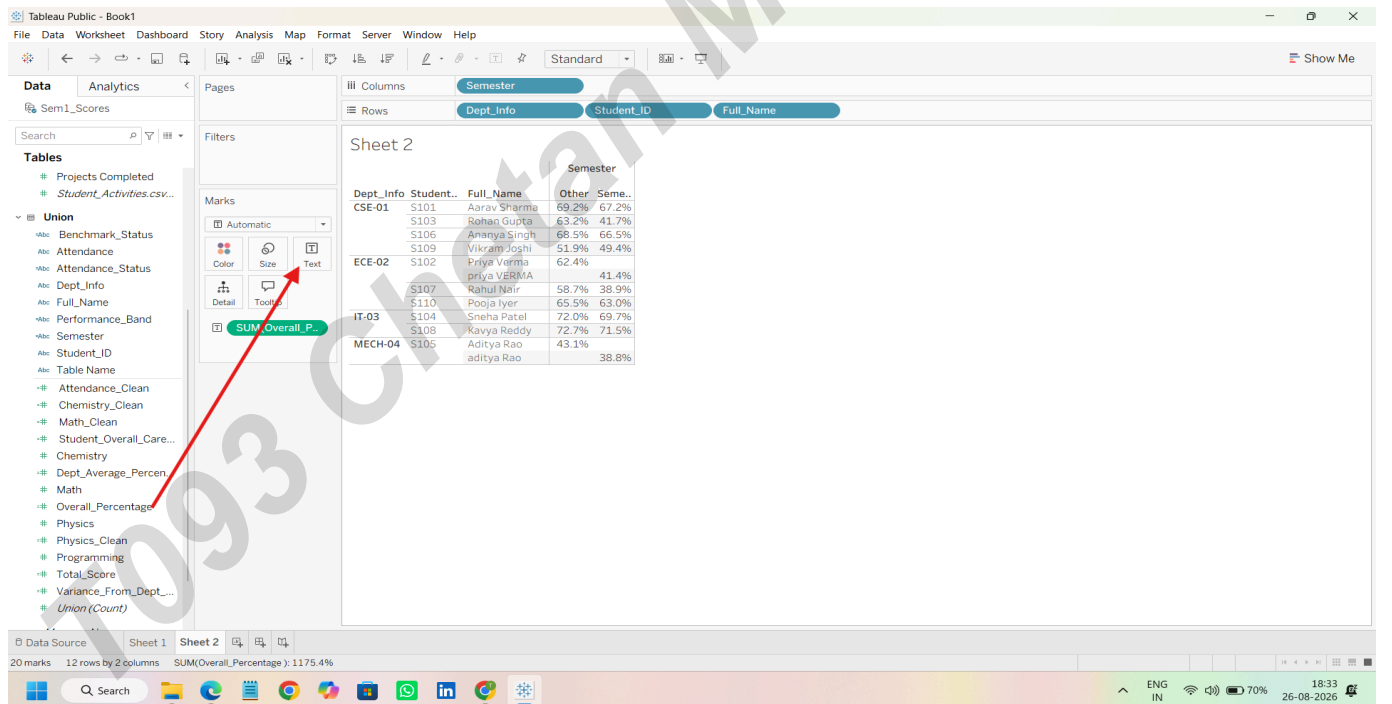
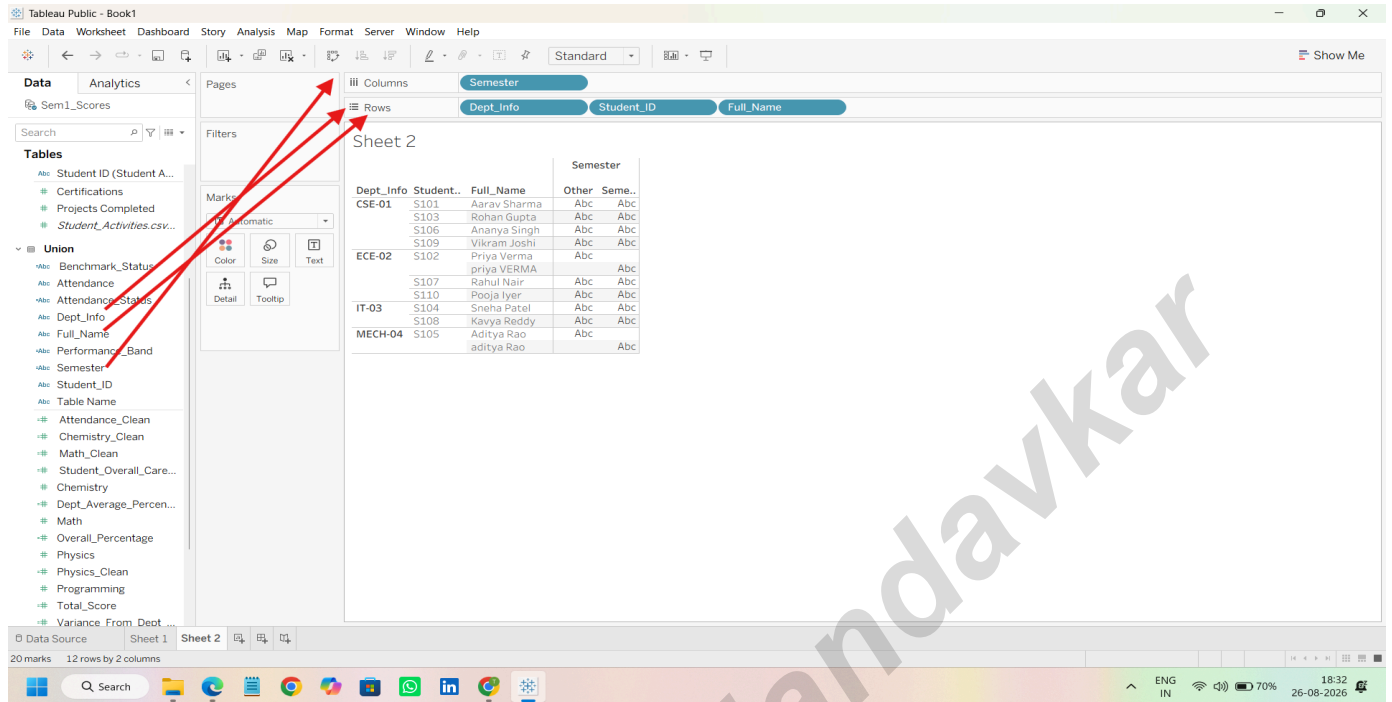
1. Drag Dept_Info, Student_ID, and Full_Name to the Rows shelf.
2. Drag Semester to the Columns shelf.
3. Drag Overall_Percentage onto Text on the Marks card.
4. Drag Dept_Average_Percentage_LOD onto Tooltip on the Marks card.
5. Drag Benchmark_Status onto Color on the Marks card.

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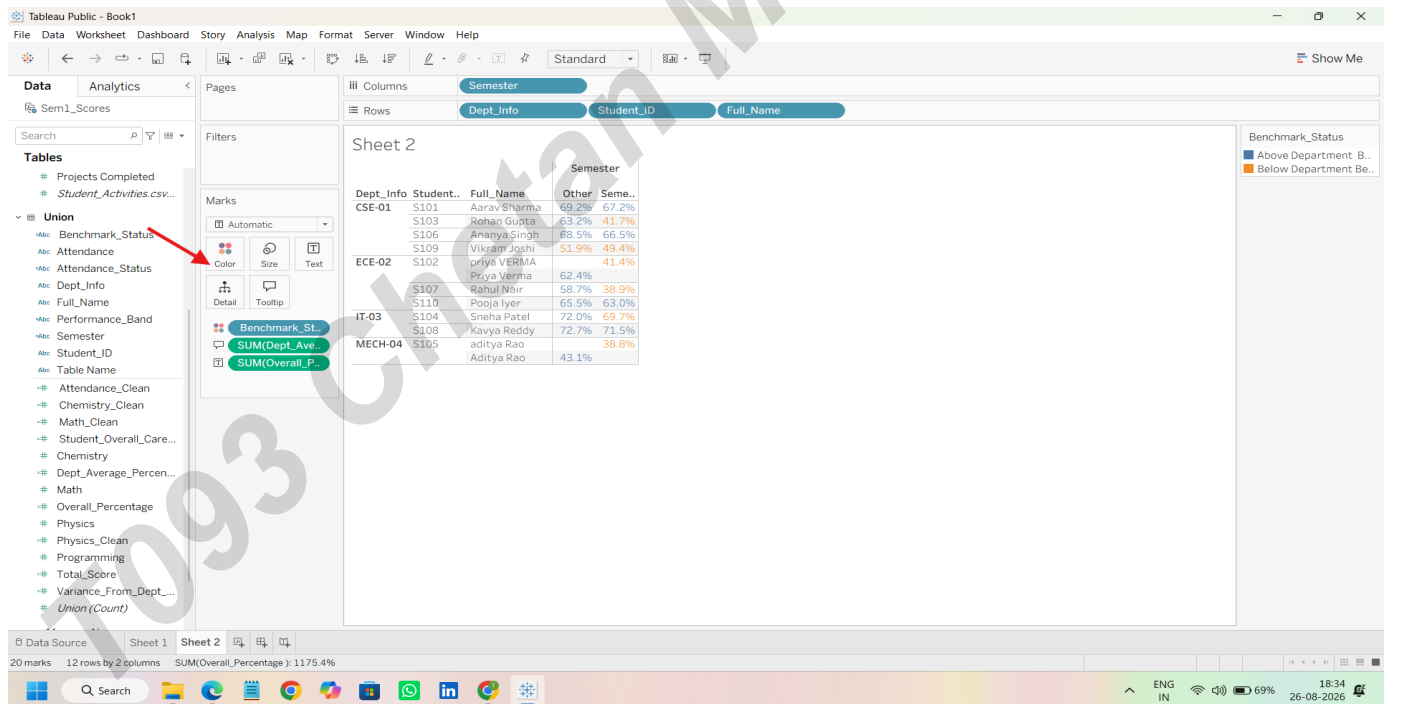
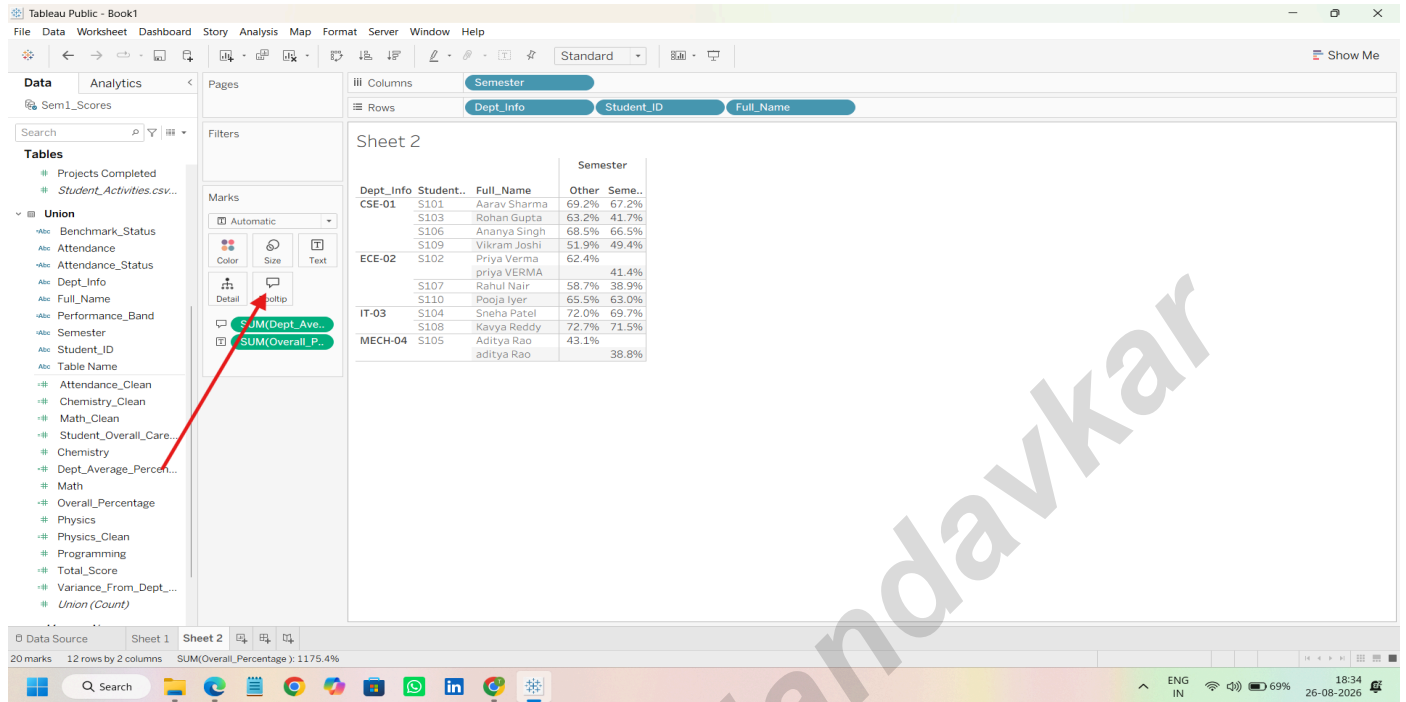


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G: Apply Advanced Data Cleaning Techniques

Step 1:- Navigate to Sheet 1 (or a new worksheet).

2. Right-click an empty space in the Data Pane --> select Create Calculated Field....

3. Create the trimmed and standardized student name field:

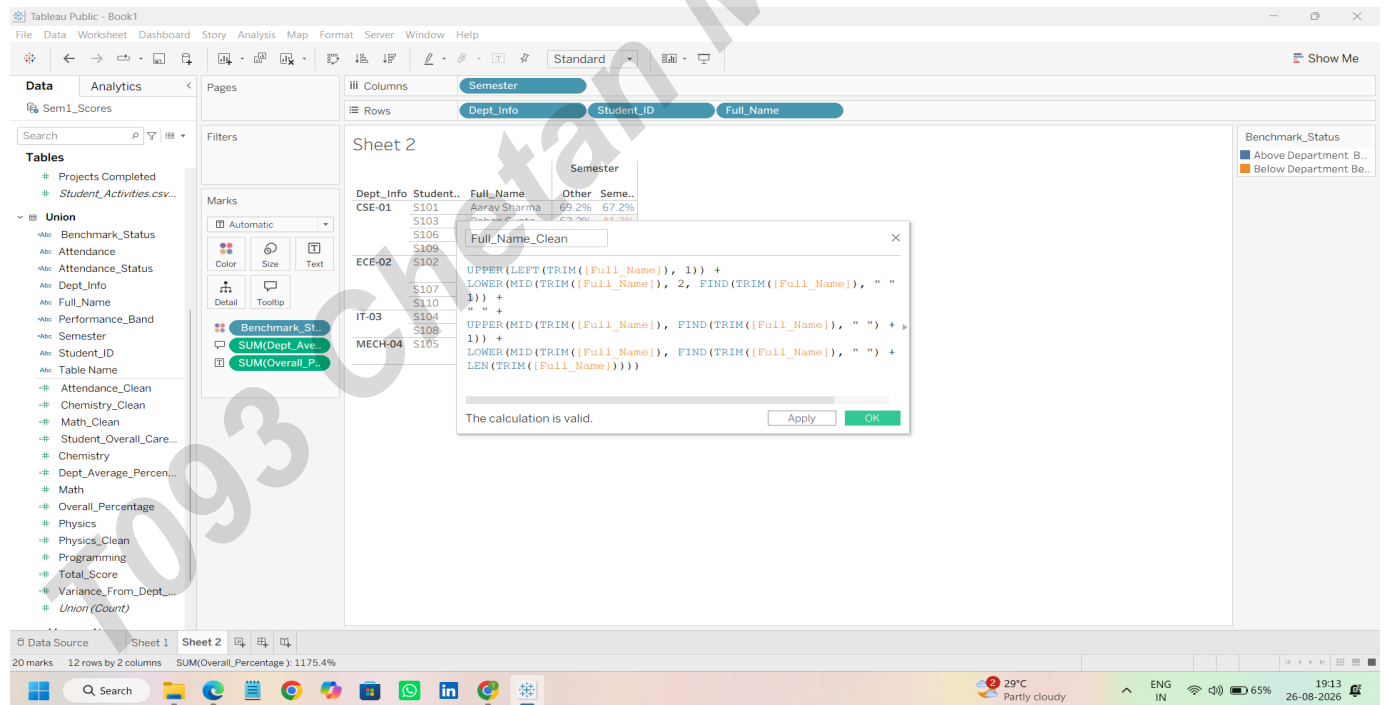
- o Field Name: Full_Name_Clean

- o Tableau Formula:

```
Code snippet UPPER(LEFT(TRIM([Full_Name]), 1)) +  
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") - 1)) +  
" " + UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1,  
1)) + LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") +  
2, LEN(TRIM([Full_Name]))))
```

- o (Alternatively, for standard trim-only cleaning): Code snippet
TRIM([Full_Name])

- o Click OK.



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Step 2:- Extract the First Name:

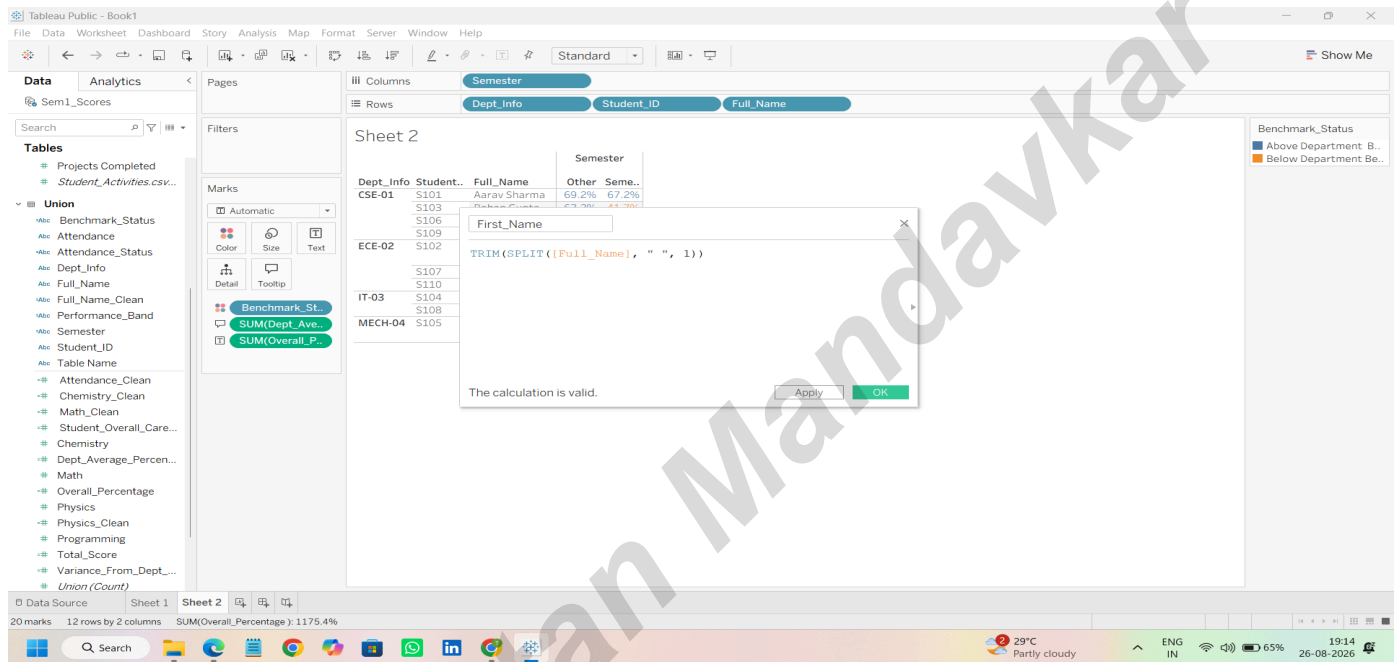
o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: First_Name

o Tableau Formula:

Code snippet `TRIM(SPLIT([Full_Name], " ", 1))`

o Click OK.



Step 3:- Extract the Last Name:

o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: Last_Name

o Tableau Formula:

Code snippet `TRIM(SPLIT([Full_Name], " ", -1))`

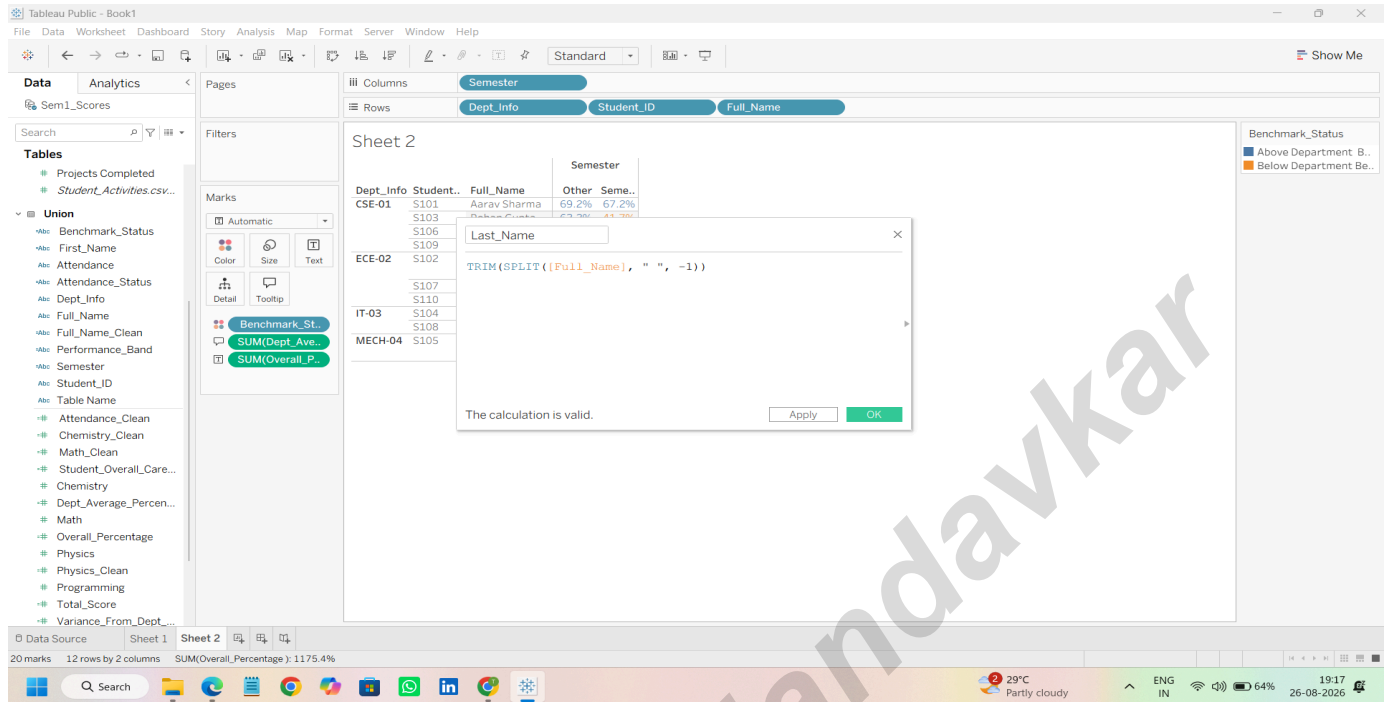
o Click OK.

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Step 4:- Extract the Department Acronym:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Dept_Branch
- o Tableau Formula:
Code snippet `SPLIT([Dept_Info], "-", 1)`
- o Click OK.

2. Extract the Batch / Section ID:

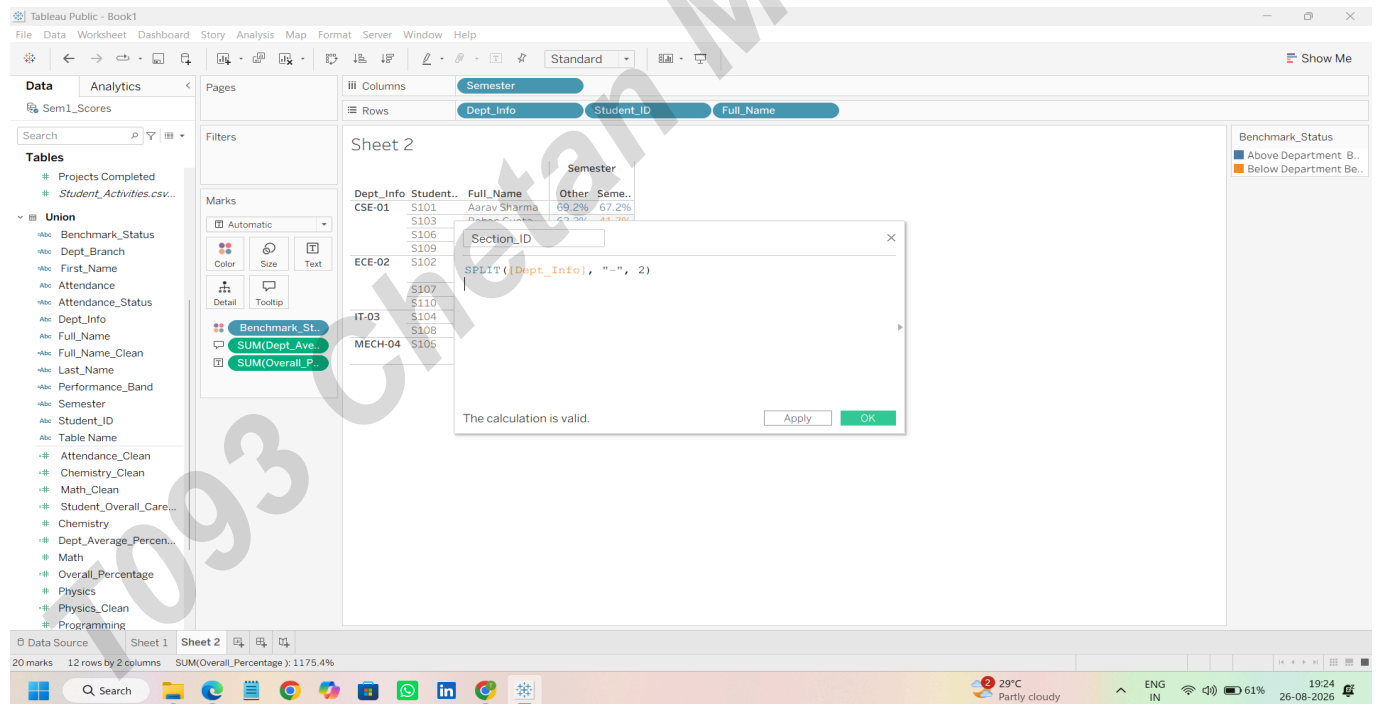
- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Section_ID
- o Tableau Formula: Code snippet `SPLIT([Dept_Info], "-", 2)`

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The screenshot displays the Tableau Desktop environment. On the left, the 'Tables' pane lists various data sources, including 'Sem1_Scores' and 'Union'. The 'Marks' pane shows the current visualization type as 'Automatic'. The main view is a data table with columns: Dept_Info, Student_ID, Full_Name, and Semester. A dialog box is open for editing the 'Dept_Branch' field, showing a formula: `SPLIT([Dept_Info], "-", 1)`. The interface includes a sidebar with 'Tables' and 'Marks' sections, and a top menu bar with 'File', 'Data', 'Worksheet', 'Dashboard', etc.



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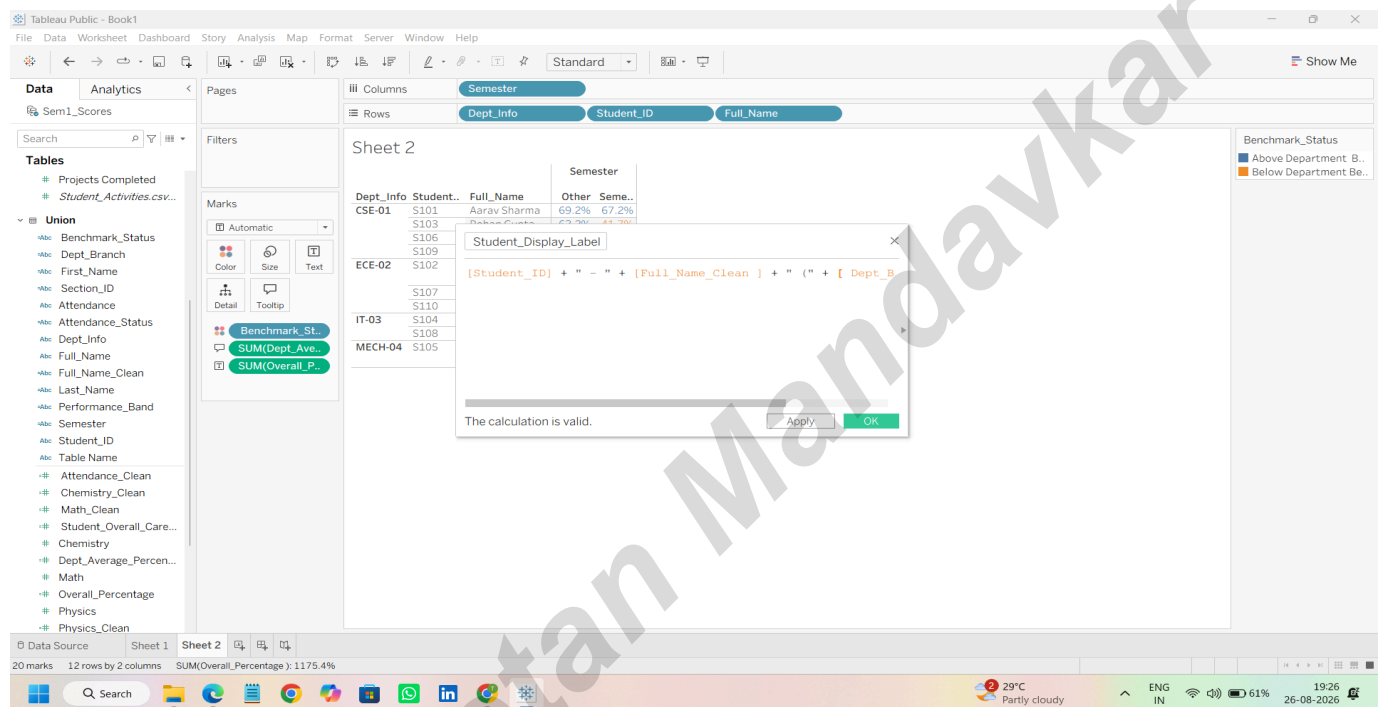
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Step 5:- Right-click in the Data Pane --> select Create Calculated Field....

2. Field Name: Student_Display_Label

3. Tableau Formula:

Code snippet [Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")" 4. Click OK.



Step 5:- 1. Place Student_Display_Label on the Rows shelf.

2. Place Semester on the Columns shelf.

3. Place Overall_Percentage on Text.

4. Place Performance_Band on Color.

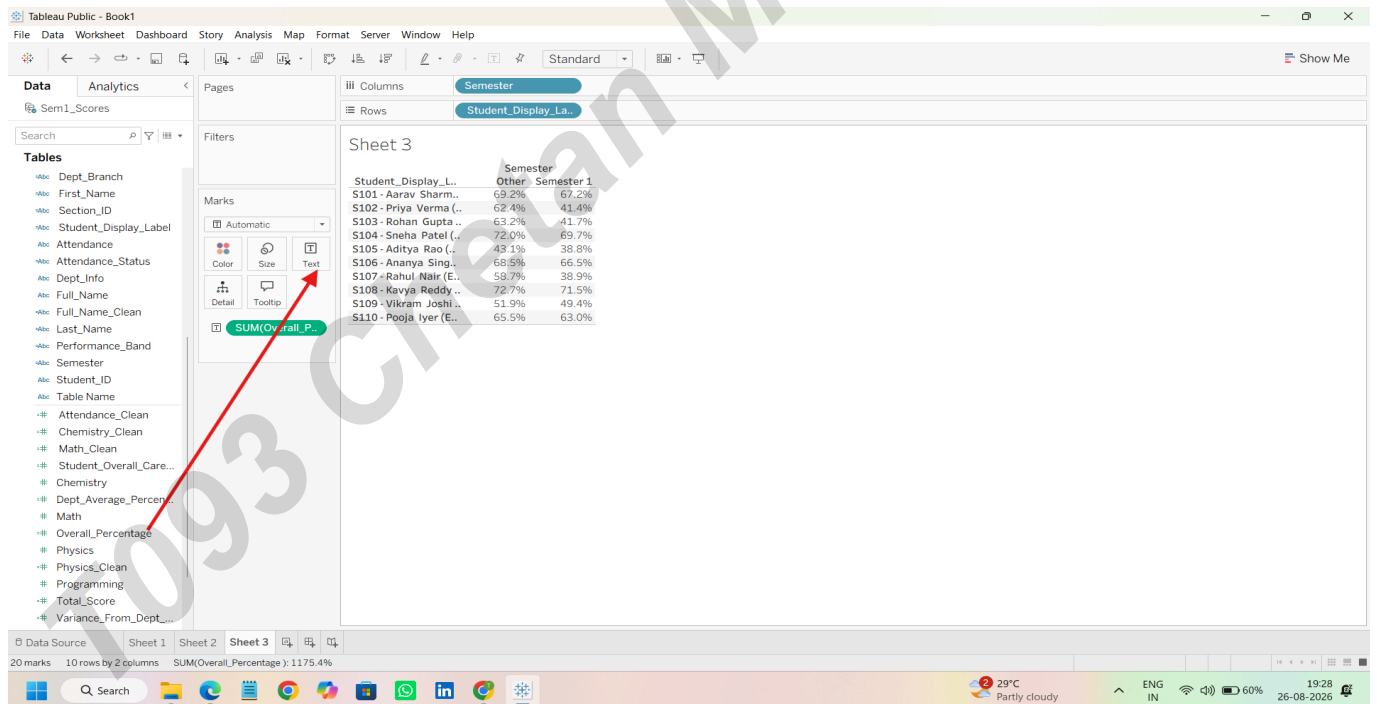
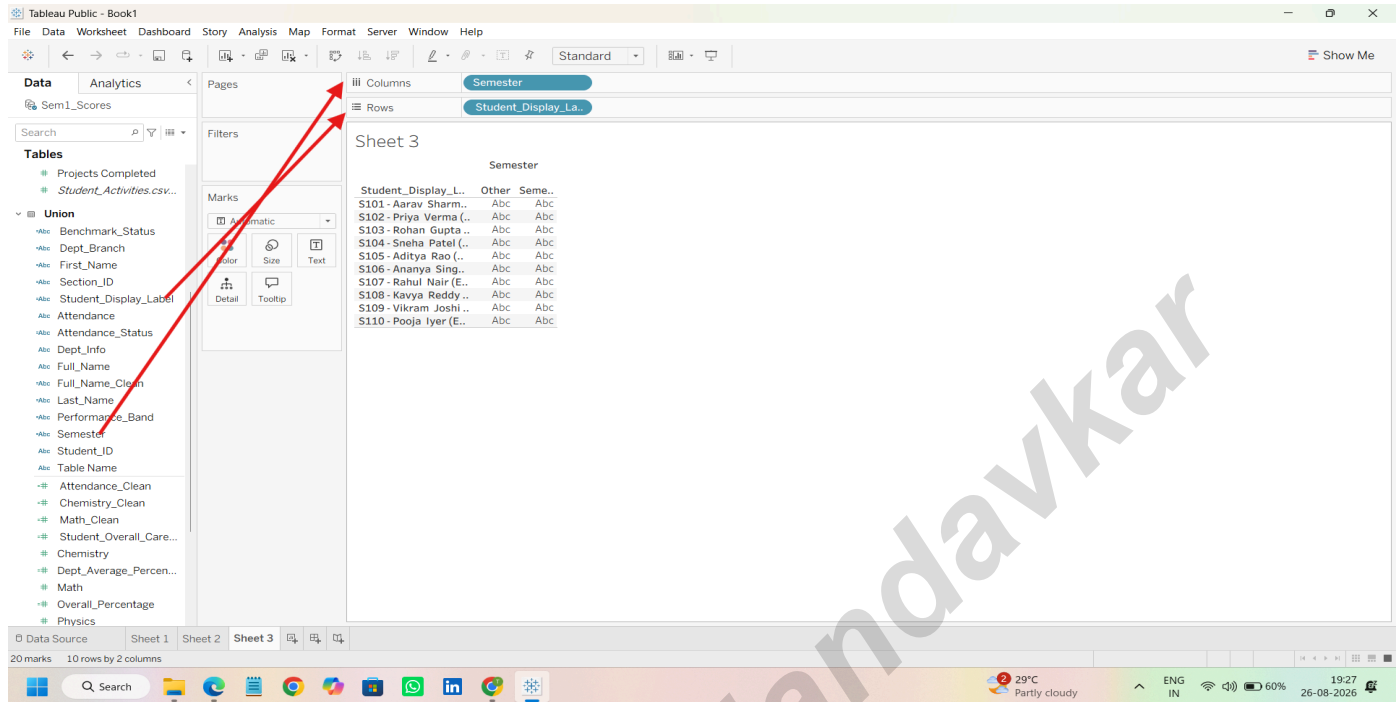
5. Place Dept_Branch and Section_ID in the Filters shelf. (While Filtering You Can select any of the value you want)

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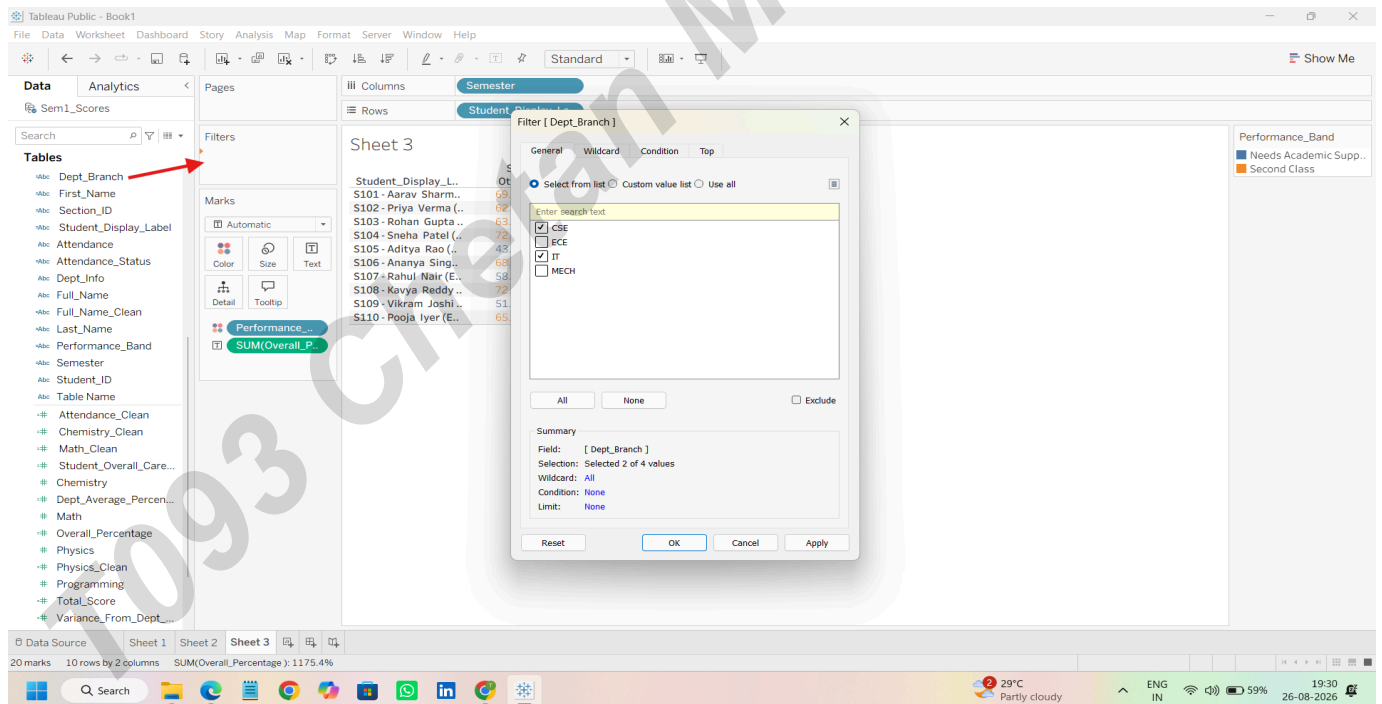
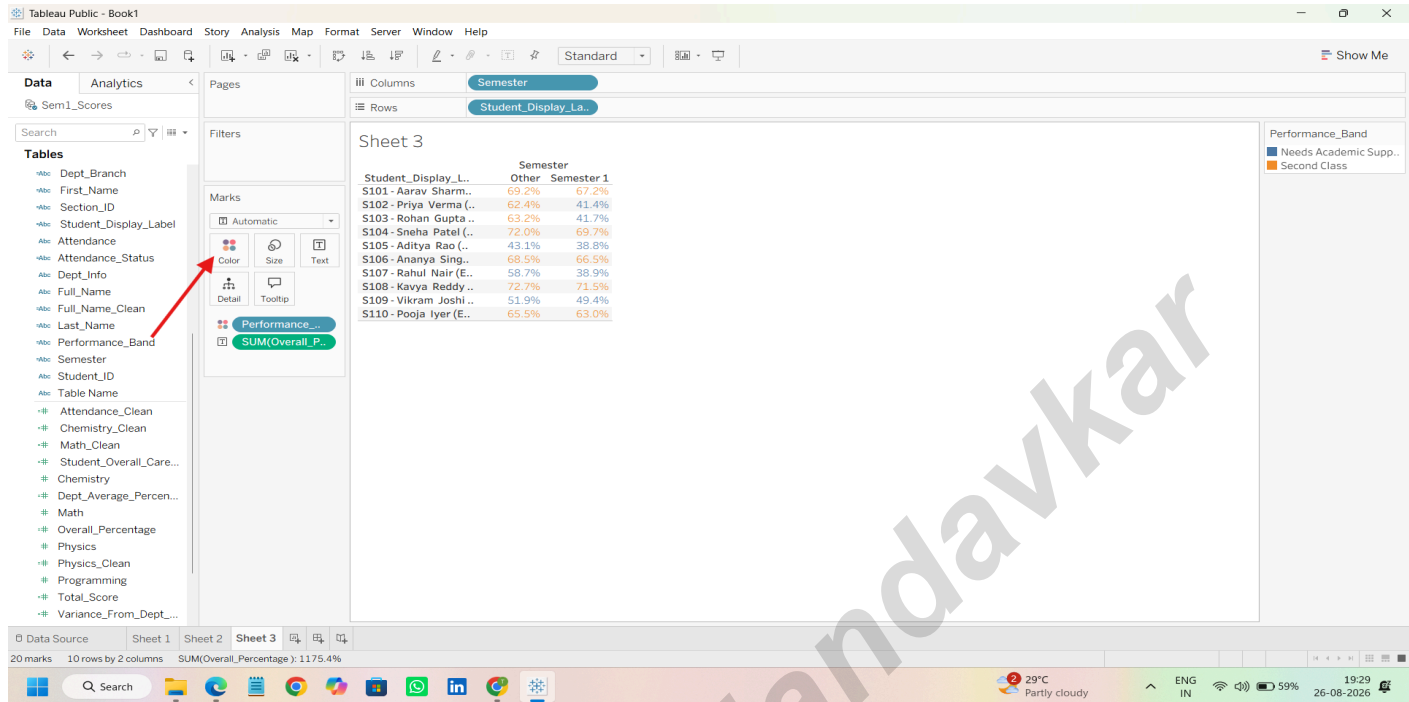


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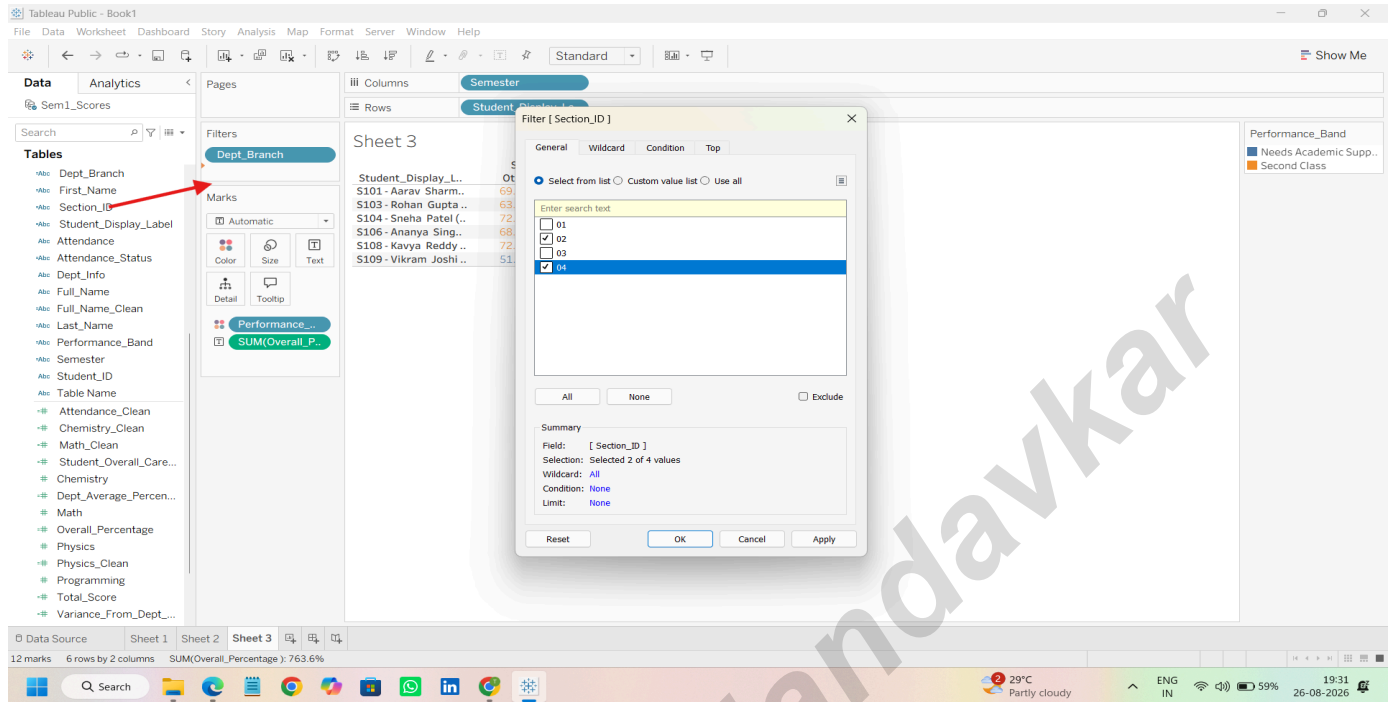


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